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EDITED BY

Bilal Üstün,
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and Information Technology, Egypt
Dimple Chawla,
Vivekananda Institute of Professional
Studies-Technical Campus, India

*CORRESPONDENCE

Aizhan Nazyrova
✉ nazyrova_aye_1@enu.kz
Zhanar Lamasheva
✉ lamasheva_zhb@enu.kz
Albina Gibadullina
✉ gibadullina_aye_1@enu.kz

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An explainable machine learning framework for academic stress classification among university students: a comparative multi-dataset study with ablation analysis and statistical significance testing

Assel Omarbekova, Aizhan Nazyrova*, Gulmira Bekmanova,
Zhanar Lamasheva*, Albina Gibadullina* and Assem Abdikarim

Institute of Digital Sciences and Artificial Intelligence, L.N. Gumilyov Eurasian National University,
Astana, Kazakhstan

Academic stress among university students represents a major public health concern due to its detrimental effects on cognitive functioning, psychological well-being, and long-term academic performance. Although machine learning has increasingly been applied to stress prediction, existing studies commonly rely on single datasets, insufficiently address class imbalance, and provide limited model interpretability. This study proposes a comparative machine learning framework that integrates explainable artificial intelligence (XAI), ablation analysis, and statistical significance testing to improve the classification of student stress. Two complementary datasets were analysed. Dataset 1 included 777 university students aged 18–22 years with three stress-type categories (Eustress, Distress, and No Stress), while Dataset 2 comprised 1,100 records with 20 psychological, physiological, environmental, and academic features labelled according to three stress severity levels (Low, Medium, and High). Five supervised learning algorithms—Logistic Regression, Random Forest, Gradient Boosting, Support Vector Machine with a radial basis function kernel, and Multilayer Perceptron—were evaluated using 10-times repeated stratified 5-fold cross-validation. Severe class imbalance in Dataset 1 (91.4% Eustress) was addressed using the Synthetic Minority Over-sampling Technique (SMOTE), and its impact was quantified through a structured ablation study. Model performance differences were assessed using the Friedman test with post-hoc Nemenyi analysis, and SHAP was employed to provide global and class-specific model interpretability. The Friedman test demonstrated statistically significant differences among classifiers ($\chi^2 = 68.97$, $*p < 0.001$), with post-hoc analysis identifying Random Forest and Logistic Regression as the statistically superior-performing models. Random Forest achieved the highest overall classification accuracy (0.8909) and the lowest log loss (0.2057), whereas Logistic Regression produced the highest discriminative performance (AUC-ROC = 0.9853). The ablation study confirmed that SMOTE substantially improved minority-class prediction performance. SHAP analysis identified blood pressure, sleep quality, teacher–student relationship, and academic performance as the most influential predictors of overall stress severity, while self-esteem, sleep quality, bullying, and anxiety level were the strongest contributors to the High Stress class. The proposed framework provides a reproducible, interpretable, and

statistically validated approach for multi-class stress classification among university students. By combining class imbalance correction, rigorous comparative evaluation, explainable artificial intelligence, and statistical validation, the framework addresses important methodological limitations of previous studies and offers a robust foundation for the development of early stress detection and intervention systems in higher education.

KEYWORDS

academic stress, class imbalance, explainable artificial intelligence, machine learning, Random Forest, SHAP, stress classification, student mental health

1 Introduction

The mental health of university students has emerged as one of the most pressing concerns in contemporary higher education. According to the World Health Organization, depressive and anxiety disorders affect hundreds of millions of individuals globally, with young people constituting a disproportionately vulnerable demographic (World Health Organization, 2025a,b; UNICEF, 2021). The National Institute of Mental Health estimates that nearly one in five adults experiences a mental illness in any given year, a burden that falls particularly heavily on those navigating the compounding demands of academic environments (National Institute of Mental Health, 2023). International organisations have further emphasised that learning outcomes cannot be sustainably improved without addressing the psychological conditions under which students operate (OECD, 2023; UNESCO, 2023).

Academic stress occupies a central position within this broader mental health landscape. Unlike generalised psychological distress, academic stress is embedded in the specific ecological conditions of university life: examination pressure, competitive peer dynamics, perceived parental expectations, insufficient sleep, and uncertainties about future career trajectories. Neuroscience research has demonstrated that chronic academic stress impairs working memory and attentional control, thereby directly undermining the cognitive processes on which academic achievement depends (Almarzouki, 2024). Epidemiological studies further confirm that a substantial proportion of students' experience moderate-to-high stress levels throughout their enrolment, with negative downstream effects on psychological wellbeing and long-term educational outcomes (Campbell et al., 2022; Córdova Olivera et al., 2023).

Contemporary psychological theory distinguishes between eustress—motivationally adaptive stress that facilitates goal pursuit and performance—and distress, characterised by maladaptive arousal, cognitive disruption, and impaired coping (Fullerton et al., 2021). This distinction carries significant practical implications for intervention design: eustress may not warrant clinical concern and may even be cultivated, whereas distress requires timely identification and targeted support. A classification framework capable of differentiating stress types and severities is therefore not merely an academic exercise but a prerequisite for effective institutional response.

Traditional approaches to student stress assessment rely predominantly on standardised self-report instruments such as the Perceived Stress Scale or the Depression Anxiety Stress Scales. While validated, these instruments are inherently retrospective and are not well suited for continuous population-scale monitoring (Mittal et al., 2022; Daza et al., 2023). The expansion of digital data collection within educational institutions has consequently created new opportunities for

data-driven approaches to student welfare monitoring (Liu et al., 2025; Mittal et al., 2022).

The machine learning literature on student stress has grown substantially in recent years, encompassing physiological signal processing using EEG and photo plethysmography (AlShorman et al., 2022; Anwar and Zakir, 2022; Fernandez et al., 2025), text-based social media analysis (Oryngozha et al., 2024), and sensor fusion from wearable devices (Luo et al., 2024; Vos et al., 2023a,b). Despite demonstrating competitive predictive performance, these approaches share several limitations from a translational perspective. First, the majority of existing studies evaluate a single algorithm on a single dataset, precluding meaningful conclusions about which model characteristics generalise across stress conceptualisations (Daza et al., 2023; Liu et al., 2025). Second, the dominant focus on predictive accuracy has marginalised questions of model interpretability (Tariq et al., 2025). Third, class imbalance represents a pervasive challenge that has received uneven treatment, with standard classifiers exhibiting systematic bias toward majority classes (Zhang et al., 2022). Fourth, most studies do not apply statistical significance testing to determine whether reported performance differences among competing algorithms are distinguishable from chance variation.

This study addresses these limitations by proposing a comparative machine learning framework for the classification of academic stress among university students. The scientific contribution lies not in the introduction of new individual algorithms, but in the design and empirical validation of an integrated analytical framework that simultaneously addresses four limitations absent from any single prior study: dual-dataset evaluation across two distinct stress conceptualisations, structured ablation analysis to isolate the contribution of each pipeline component, non-parametric statistical significance testing to validate all performance comparisons, and SHAP-based explainability to ensure institutional interpretability. As confirmed by the systematic review presented in Table 1, no prior study in this domain has combined all four of these components within a single reproducible framework. To guide the study, the following research questions are formulated:

RQ1: Which machine learning model provides the most accurate and statistically distinguishable prediction of student stress levels?

RQ2: How do psychological, social, physiological, and academic factors jointly influence student stress severity?

RQ3: Which individual features contribute most to stress prediction, as identified through SHAP analysis?

TABLE 1 Summary of representative approaches in student stress detection using machine learning.

Reference	Model type	Data type	Dataset	Performance	Key findings	Limitations	Research gap
AlShorman et al. (2022)	ML (SVM, KNN)	EEG	Lab-based	Acc. ~91%	Frontal EEG features effective	Controlled lab setup	Low real-world applicability
Chen and Lee (2023)	DL (CNN, LSTM)	EEG + Behavioural	Task-based	High accuracy	Cognitive load correlates with stress	Artificial environment	Limited generalisation
de Filippis and Foysal (2024)	ML (RF, GB, XGB)	Survey	Student sample	Competitive Acc.	Identifies academic stress factors	Self-reported data	Reliability bias
Fonda et al. (2024)	Ensemble (Stacking)	Academic data	Higher education	~95% AUC	Stacking improves stability	Static snapshot	No temporal modelling
Luo et al. (2024)	DL (Branched LSTM)	Mobile sensors	Real-world	AUC ~ 0.87	Personalised prediction	Sensor dependency	Deployment challenges
Naegelin et al. (2023)	Interpretable ML	Multimodal	Simulated office	Competitive	Interpretable multimodal detection	Task-specific	Generalisation needed
Ratul et al. (2023)	ML (LR, RF, SVM)	Survey	University	Acc. ~82%	Social/psychological factors	Single dataset	No cross-dataset validation
Tariq et al. (2025)	XAI + ML (RF, XGB)	Survey	Higher education	High Acc. + SHAP	XAI improves transparency	Limited diversity	Interaction factors needed
Liu and Yu (2026)	Interpretable ML (DT)	Stress data	Student dataset	Competitive	Decision-tree explainable	Simpler model misses patterns	Need multimodal extensions
Ahmed et al. (2026)	Ensemble + XAI	Survey	University	High Acc.	Custom ensemble improves interpretability	Dataset-specific	Cross-dataset evidence needed

RQ4: How stable and generalisable are the trained models under repeated cross-validation, and are performance differences between models statistically significant?

2 Literature review

Academic stress is among the most consistently documented psychological phenomena in the university student literature. Large-scale surveys conducted across North America, Europe, and Asia have established that between 40 and 80% of university students report clinically meaningful levels of stress during their studies, with prevalence rates elevated further during examination periods and in the post-pandemic context ([Abelson et al., 2022](#); [Campbell et al., 2022](#); [Gasser et al., 2025](#)). The OECD's analysis of PISA 2022 data confirms that elevated academic anxiety and stress are associated with lower learning outcomes across educational systems ([OECD, 2023](#)).

The consequences of unmanaged academic stress extend across multiple domains. Cognitively, chronic stress impairs prefrontal cortical networks responsible for working memory, executive function, and attentional control ([Almarzouki, 2024](#)). Physiologically, prolonged stress activation produces neuroendocrine dysregulation, sleep disruption, somatic symptoms, and compromised immune function ([Figueroa et al., 2023](#)). The distinction between eustress and distress, first formalised by Selye and subsequently elaborated within cognitive appraisal frameworks, is clinically and practically significant ([Fullerton et al., 2021](#)). A classification system capable of distinguishing these qualitatively different stress states provides considerably more actionable information than a simple binary prediction.

The application of machine learning to stress detection has expanded substantially in recent literature. Physiological signal-based approaches using EEG have demonstrated real-time stress classification with accuracies exceeding 90% in laboratory conditions ([AlShorman et al., 2022](#); [Fernandez et al., 2025](#); [Hafeez and Shakir, 2024](#)), while photoplethysmography-based approaches using heart rate variability have shown comparable results ([Anwar and Zakir, 2022](#); [Nazeer et al., 2024](#)). However, the controlled nature of these experiments substantially limits their ecological validity and institutional scalability. Survey and academic data-based approaches are more directly applicable to university settings, with ensemble tree methods including Random Forest and Gradient Boosting reported to achieve accuracies ranging from 80 to 95% ([Fonda et al., 2024](#); [Ratul et al., 2023](#); [Tariq et al., 2025](#)). Systematic reviews of the field confirm that ensemble methods consistently achieve competitive performance on structured educational datasets while maintaining computational efficiency and interpretability advantages over deep architectures ([Daza et al., 2023](#); [Liu et al., 2025](#)).

The deployment of machine learning models in educational settings has increasingly been conditioned on their interpretability. Shapley Additive Explanations (SHAP), grounded in cooperative game theory, provide a mathematically rigorous framework for attributing model predictions to individual input features ([Gupta et al., 2023](#)). Unlike global feature importance measures such as impurity-based rankings, SHAP values are additive and consistent. The TreeSHAP algorithm computes exact Shapley values for tree-based models in polynomial time ([Liu and Yu, 2026](#)), and SHAP analyses in the student stress domain have consistently revealed that academic performance, sleep quality, and social support variables emerge as high-importance predictors ([Ahmed et al., 2026](#); [Tariq et al., 2025](#)).

Class imbalance is a pervasive challenge in student mental health datasets, reflecting genuine distributional asymmetry in the underlying population. The Synthetic Minority Over-sampling Technique (SMOTE) addresses this challenge by generating synthetic minority-class samples through linear interpolation between existing observations and their k -nearest neighbours in feature space (de Filippis and Foysal, 2024; Tseng et al., 2024). A critical methodological gap in the existing literature concerns the absence of statistical significance testing for reported performance differences. Without non-parametric tests such as the Friedman test and *post-hoc* Nemenyi analysis, it is impossible to determine whether observed differences between classifiers reflect genuine algorithmic superiority or sampling variability.

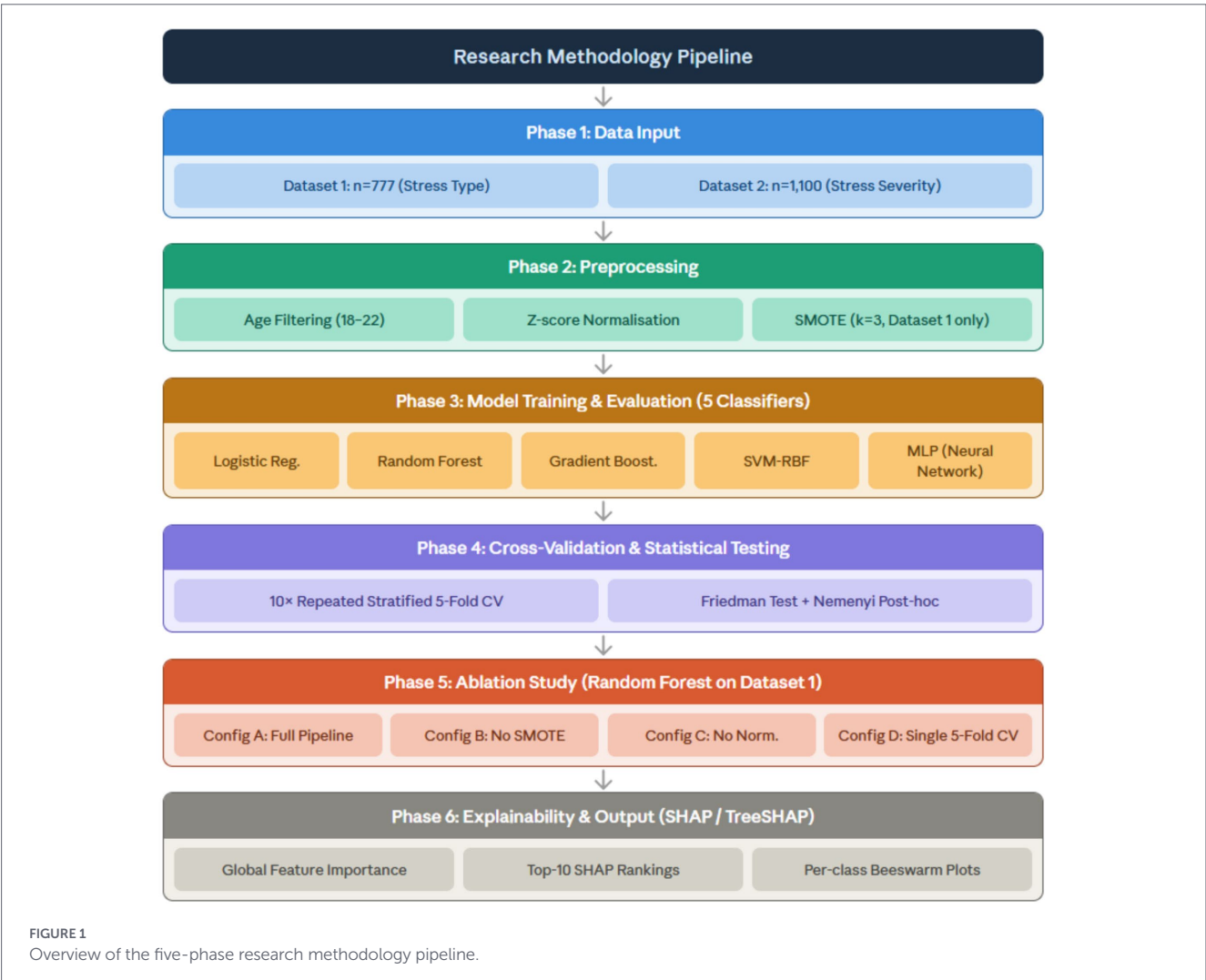
The analysis presented in Table 1 confirms three recurring gaps in the existing literature—absent multi-dataset evaluation, inconsistent imbalance correction, and lack of statistical significance testing—which the present study is designed to address.

Recent studies published between 2024 and 2026 have further advanced the field of explainable AI and educational mental health analytics. Tariq et al. (2025) applied explainable AI methods combined with machine learning to student stress prediction in higher education, directly supporting the methodological approach adopted in the present study. Ahmed et al. (2026) proposed a custom ensemble model combining hyperparameter optimisation with explainability

techniques for predicting mental health among university students. Liu and Yu (2026) developed an interpretable decision tree approach for student stress prediction, emphasising the value of model transparency for educational practitioners. Duan (2026) employed fuzzy logic methods for student mental health assessment, highlighting the growing diversity of interpretable modelling approaches in this domain. Collectively, these recent contributions confirm that the integration of explainability methods with robust statistical validation—as implemented in the present framework—represents the current frontier of the field.

3 Methods

The study is implemented as a five-phase pipeline, illustrated in Figure 1. Each phase addresses a distinct methodological challenge: Phase 1 preprocesses and partitions the two datasets; Phase 2 trains and evaluates five classification algorithms across eight performance metrics; Phase 3 applies repeated stratified cross-validation followed by the Friedman test and Nemenyi post-hoc analysis to establish statistical significance; Phase 4 conducts a structured ablation study to isolate the contribution of individual pipeline components; and Phase 5 applies TreeSHAP-based explainability to



identify the principal predictors of student stress. Two datasets serve as inputs throughout: an imbalanced survey Dataset 1 ($n = 777$) and a balanced Dataset 2 ($n = 1,100$). The complete step-by-step specification of the pipeline is formalised in [Algorithm 1](#).

The pipeline is structured to address three interrelated methodological challenges simultaneously. First, the dual-dataset design separates two fundamentally different classification problems – stress type identification (Dataset 1) and stress severity grading (Dataset 2)—ensuring that findings are not conflated across conceptually distinct operationalisations of academic stress. Second, the sequential dependency structure of phases ensures that no information from the test partition can influence preprocessing parameters, model selection, or cross-validation scores, thereby guaranteeing unbiased generalisation estimates. Third, the ablation and statistical testing phases are embedded as first-class analytical components rather than post-hoc additions, making it possible to draw rigorous causal conclusions about the contribution of each pipeline component and the reliability of each reported performance difference.

3.1 Datasets

Two complementary datasets were employed to examine academic stress from distinct conceptual perspectives, enabling a more comprehensive evaluation than would be possible with a single data source.

Dataset 1 (Stress_Dataset.csv) is publicly available under the title “Student Stress Monitoring Datasets” at: https://www.kaggle.com/datasets/mdsultanulislamovi/student-stress-monitoring-datasets?select=Stress_Dataset.csv. The dataset captures survey responses from 843 college students aged 18–22, collected via Google Forms using a five-point Likert scale ranging from “Not at all” to “Extremely,” and anonymised prior to publication. The questionnaire comprised 23 items covering emotional, physical, academic, environmental, and social stress indicators. The complete list of questionnaire items is provided in [Appendix A](#). The original dataset comprised 843 responses; age filtering to the 18–22 year range yielded 777 valid records. The target variable reflected a

ALGORITHM 1 Overall stress classification framework.

Input: Dataset 1 ($n=777$, stress type labels) ; Dataset 2 ($n=1100$, stress severity labels)

1 Phase 1 — Data Input

2 Phase 2 — Preprocessing

1 Filter Dataset 1 to age range [18, 22] ; retain $n = 777$ records

2 Apply stratified 80/20 train-test split to Dataset 1 and Dataset 2

3 Fit z-score parameters (μ , σ) on training partitions only

4 Transform: $z_i = (x_i - \mu) / \sigma$ for all features

5 Apply SMOTE ($k=3$) to Dataset 1 training partition only

3 Phase 3 — Model Training and Evaluation

6 for each classifier $C \in \{LR, RF, GB, SVM-RBF, MLP\}$ do

6.1 Train C on Dataset 2 training partition ($n = 880$)

6.2 Predict on Dataset 2 test partition ($n = 220$)

6.3 Compute 8 metrics: Acc, Prec, Rec, F1, AUC-ROC, MCC, κ , LogLoss

4 Phase 4 — Cross-Validation and Statistical Testing

7 Run 10x5 Repeated Stratified K-Fold CV on Dataset 2 training partition

8 Apply Friedman test to S ; compute χ^2 and p-value

9 if $p < 0.05$: run post-hoc Nemenyi pairwise analysis

5 Phase 5 — Ablation Study

10 Evaluate RF on Dataset 1 under 4 configurations :

Config A: Full pipeline (SMOTE + normalisation + 10x5 RSKF)

Config B: Without SMOTE (imbalanced training)

Config C: Without normalisation (SMOTE only)

Config D: Single 5-fold CV (no repetition)

6 Phase 6 — Explainability

11 Apply TreeSHAP to fitted RF on Dataset 2 test partition

12 Compute global importance : $\text{mean } |\phi_i|$ across instances and classes

13 Rank top-10 features ; generate per-class beeswarm plots

14 return classifier profiles, cluster structure, top-10 SHAP ranking

Output: Classifier performance profiles ; Ablation results ; Significance clusters ; SHAP rankings

three-class stress type label (Eustress, Distress, No Stress) based on the theoretical distinction between adaptive and maladaptive stress responses. The pre-filtering class distribution revealed a severe imbalance: 91.4% Eustress, 4.9% No Stress, and 3.7% Distress, necessitating oversampling prior to model training.

Dataset 2 (StressLevelDataset.csv) is publicly available under the title “Student Stress Factors” at: <https://www.kaggle.com/datasets/samyakb/student-stress-factors>. The dataset comprises 1,100 student records with 20 features measured on a self-report ordinal scale from 1 to 5, grouped across five scientifically recognised domains: psychological (anxiety_level, self_esteem, mental_health_history, depression), physiological (headache, blood_pressure, sleep_quality, breathing_problem), environmental (noise_level, living_conditions, safety, basic_needs), academic (academic_performance, study_load, teacher_student_relationship, future_career_concerns), and social (social_support, peer_pressure, extracurricular_activities, bullying). The target variable was a three-class stress severity label: Low (0), Medium (1), and High (2). The dataset exhibited a near-balanced distribution (33.9% Low, 32.5% Medium, 33.5% High), making it well suited for comparative algorithm evaluation without additional oversampling. Dataset 2 served as the primary evaluation dataset for all classification experiments.

Both datasets are fully anonymised and publicly available. No primary data collection was conducted by the authors of the present study. As both datasets are publicly accessible, contain no personal identifiers, and were obtained exclusively for research purposes in accordance with the respective platform terms of use, no institutional ethical approval was required for this study.

A structured comparison of the two datasets is provided in Table 2. The class distributions before and after SMOTE application are illustrated in Figure 2.

3.2 Data preprocessing

Both datasets were verified to contain no missing values. Dataset 1 was restricted to students aged 18–22 to ensure age-group homogeneity. Categorical symptom responses were encoded

using label encoding, and column names were standardised to concise descriptive labels.

All 20 continuous predictor features in Dataset 2 were standardised using z-score normalisation prior to model fitting, ensuring that classifiers with distance-based or gradient-based inductive biases – specifically SVM-RBF, MLP, and Logistic Regression—were not adversely affected by differences in feature scale. The standardisation transformation applied to each feature x_i is given by Equation 1:

$$z_i = (x_i - \mu) / \sigma, \text{ where } \mu = \left(\frac{1}{n} \right) \sum_i x_i, \\ \sigma = \sqrt{\left(\frac{1}{(n-1)} \right) \sum_i (x_i - \mu)^2} \quad (1)$$

Standardisation parameters were estimated exclusively from the training partition and subsequently applied to both training and test data, ensuring no information leakage from test observations.

The severe class imbalance in Dataset 1 (91.4% Eustress) was addressed by applying the Synthetic Minority Over-sampling Technique exclusively to the training partition following train–test split (de Filippis and Foysal, 2024). The complete oversampling procedure is specified in Algorithm 2. SMOTE generates synthetic minority-class samples by identifying the k nearest neighbours ($k = 3$) of each minority-class instance and interpolating new observations along the line segments connecting an instance to its neighbours, as defined in Equation 2:

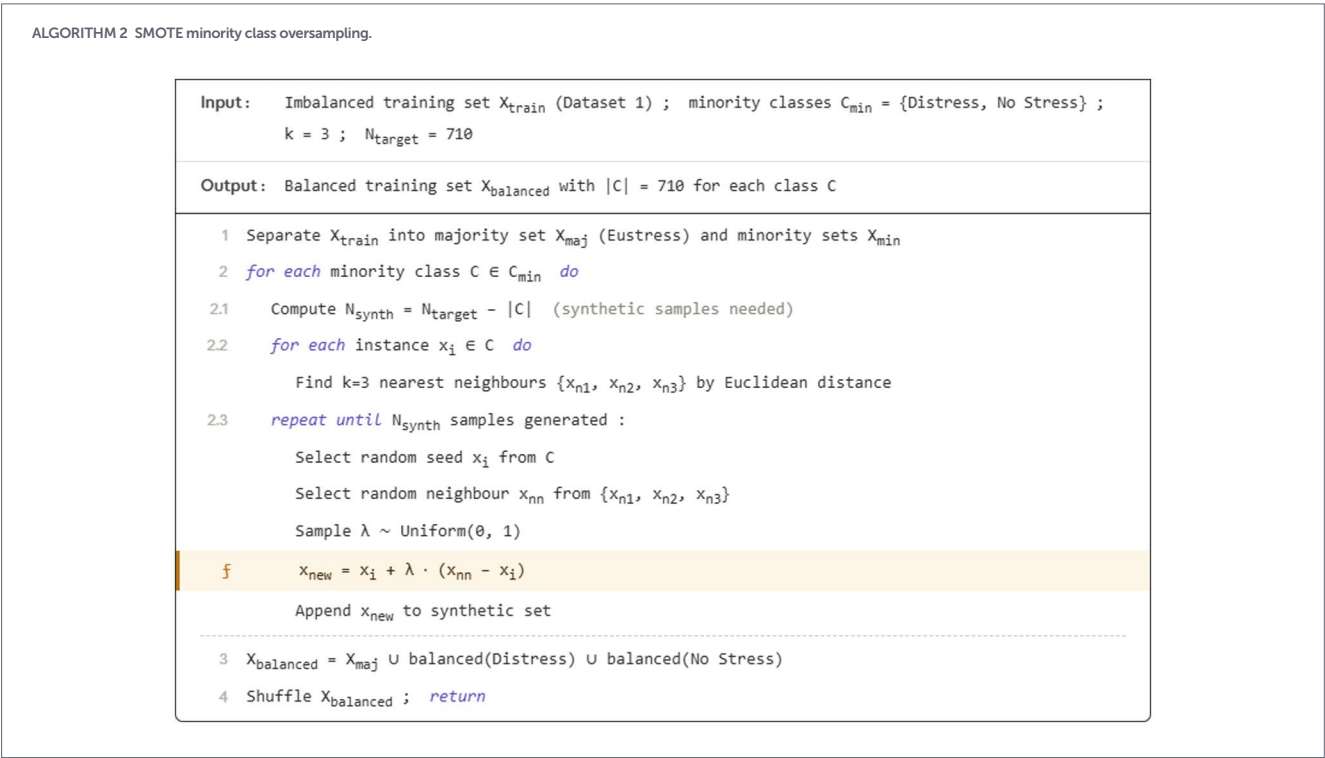
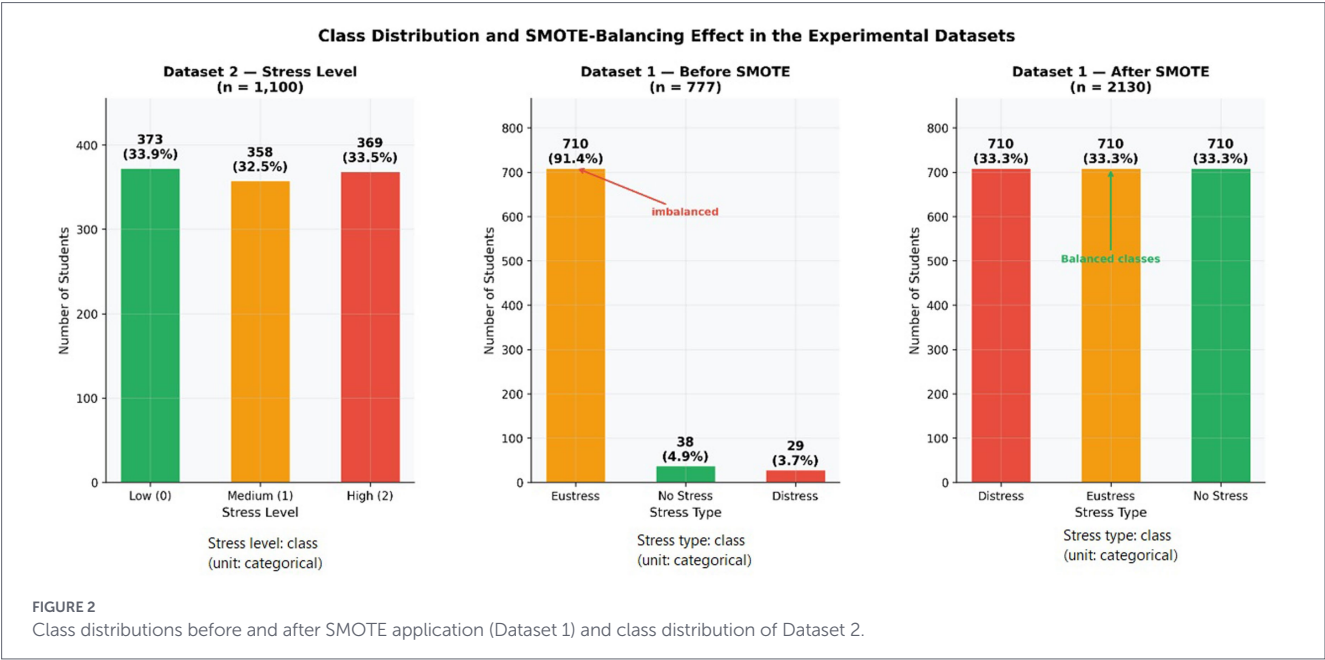
$$x_{new} = x_i + \lambda \cdot (x_{nn} - x_i), \lambda \sim \text{Uniform}(0,1) \quad (2)$$

Following SMOTE application, the training partition for Dataset 1 contained 710 observations per class (2,130 total), enabling unbiased classifier training. The test partition preserved the original class distribution to provide a realistic evaluation under the naturally occurring data distribution.

The selection of $k = 3$ nearest neighbours reflects a deliberate trade-off between two competing risks. A very small k (e.g., $k = 1$)

TABLE 2 Dataset characteristics—Dataset 1 vs. Dataset 2.

Characteristic	Dataset 1	Dataset 2
Source	Self-report questionnaire (Kaggle)	Structured student survey (Kaggle)
Total records (original)	843	1,100
Records used in study	777 (filtered: age 18–22)	1,100 (no filtering)
Number of features	23	20
Feature types	Binary (symptom indicators)	Ordinal/continuous (0–5 or 0–30)
Target variable	Stress type	Stress severity
Classes	Eustress/Distress/No Stress	Low (0)/Medium (1)/High (2)
Class distribution	Eustress: 91.4%; No Stress: 4.9%; Distress: 3.7%	Low: 33.9%; Medium: 32.5%; High: 33.5%
Class balance status	Severely imbalanced	Approximately balanced
Imbalance correction	SMOTE applied ($k = 3$, train partition only)	Not required
Training set (before SMOTE)	622 records	880 records
Training set (after SMOTE)	2,130 records (710 per class)	880 records
Test set	155 records (original distribution preserved)	220 records
Primary role in study	Stress type classification/ablation	Main evaluation dataset



generates synthetic samples in a highly localised neighbourhood, increasing the risk of producing near-duplicate instances that offer limited diversity and may create dense clusters that overfit to a narrow region of the feature space. Conversely, a large k (e.g., $k = 10$) can span boundaries between naturally occurring subgroups within a minority class, producing synthetic samples that do not faithfully represent the underlying data distribution. The value $k = 3$ is widely adopted in the SMOTE literature as a robust default that generates sufficiently diverse samples while remaining within the local neighbourhood of each seed instance (de Filippis and Foysal, 2024; Tseng et al., 2024). Crucially, because neighbours are restricted to instances of the same class, SMOTE interpolates only within the minority-class

feature space and cannot generate samples that cross the decision boundary into majority-class territory, preserving the statistical integrity of the class separation.

3.3 Classification algorithms

Five classification algorithms were selected to represent a range of inductive biases, complexity levels, and interpretability characteristics. Default hyperparameter configurations were employed throughout, as the primary objective is comparative evaluation rather than single-model optimisation. Hyperparameters for all models are presented in Table 3.

TABLE 3 Hyperparameter configurations for all five classification algorithms.

Model	Key hyperparameters	Values	Rationale/search space
Logistic Regression	Solver; regularisation; iterations	lbfgs; L2, $C = 1.0$; max_iter = 1,000; multi_class = multinomial	Standard L2 regularisation; lbfgs is numerically stable for multinomial problems; max_iter = 1,000 ensures convergence on standardised data
Random Forest	Trees; criterion; features per split	$n_estimators = 100$; Gini impurity; max_features = sqrt(p)	Breiman's original recommendation for classification: sqrt(p) features per split; 100 trees sufficient for $n < 2,000$ with diminishing returns beyond this
Gradient Boosting	Trees; learning rate; depth; loss	$n_estimators = 100$; $\eta = 0.1$; max_depth = 3; log_loss	Conservative learning rate (0.1) balances bias–variance trade-off; max_depth = 3 reduces overfitting on moderate-size datasets
SVM (RBF)	Regularisation; Kernel; γ ; probability	$C = 1.0$; RBF; $\gamma = \text{scale}$; Platt scaling for probability	Scale-normalised gamma adapts to feature variance; $C = 1.0$ provides moderate regularisation; Platt scaling required for probability output
MLP Neural Network	Architecture; activation; optimiser; α	(100, 50); ReLU; Adam; $\alpha = 0.001$; max_iter = 500	Two-layer architecture (100, 50) sufficient for 20 features; Adam optimiser standard for gradient-based training; ReLU avoids vanishing gradients
All models	Random seed	random_state = 42 for full reproducibility	Fixed random seed ensures full reproducibility of all stochastic components across experiments

The selection of five specific classifiers was motivated by a deliberate strategy of covering the full spectrum of algorithmic diversity relevant to structured survey data. Logistic Regression (LR) serves as the linear baseline: it makes the strongest distributional assumptions—namely, linear separability in the feature space—and offers the highest coefficient interpretability, making it the natural reference point against which non-linear models are compared. Random Forest (RF) and Gradient Boosting (GB) both belong to the ensemble tree family but differ fundamentally in their construction strategy: RF aggregates independently trained trees via bootstrap aggregation, reducing variance, while GB trains trees sequentially by correcting the errors of the current ensemble, reducing bias. Including both ensemble variants enables direct comparison of these two complementary bias–variance trade-off strategies. SVM-RBF introduces a kernel-based non-linear classifier that constructs an implicit high-dimensional feature mapping, offering a fundamentally different inductive bias than tree-based methods. Finally, MLP represents the neural network family, capturing feature interactions through learned hidden representations without explicit feature engineering. Together, these five classifiers span linear, bagging ensemble, boosting ensemble, kernel, and neural paradigms, ensuring that the comparative evaluation is not inadvertently biased toward any single algorithmic family. Default hyperparameter configurations were retained throughout to eliminate the confounding effect of unequal optimisation effort: any observed performance differences therefore reflect genuine algorithmic properties rather than differences in the quality of tuning.

Logistic Regression (LR) was adapted to the multiclass setting using the softmax (multinomial) formulation, modelling the posterior probability of class k given feature vector x as shown in Equation 3:

$$P(y = k | x) = \exp(w_k^T x) / \sum_j \exp(w_j^T x) \quad (3)$$

Parameters were estimated by minimising the cross-entropy loss with L2 regularisation ($C = 1.0$), as given in Equation 4:

$$\mathcal{L} = -(1/n) \sum_i \sum_K y_i^K \log(\hat{p}_i^K) + \lambda \sum_i \beta^2 \quad (4)$$

Random Forest (RF) constructs an ensemble of $T = 100$ decision trees, each trained on a bootstrap sample of the training data using a random subset of features at each split. Node splitting is governed by the Gini impurity criterion given in Equation 5:

$$\text{Gini}(t) = 1 - \sum_K p_K^2 \quad (5)$$

Classification decisions are made by majority voting across all trees, as expressed in Equation 6:

$$\hat{y} = \arg \max_k \sum_k 1[h_k(x) = k] \quad (6)$$

Gradient Boosting (GB) constructs an additive ensemble by sequentially fitting weak learners $h_m(x)$ to the pseudo-residuals of the current ensemble $F_{m-1}(x)$, as formulated in Equation 7:

$$F_m(x) = F_{m-1}(x) + \eta h_m(x) \quad (7)$$

where $\eta = 0.1$ is the learning rate. Pseudo-residuals are defined as the negative gradient of the loss, as shown in Equation 8:

$$r_{im} = -[\partial \mathcal{L}(y_i, F(x_i)) / \partial F(x_i)]|_{F = F_{m-1}} \quad (8)$$

Support Vector Machine with RBF kernel (SVM-RBF) was trained using a soft-margin formulation minimising the regularised hinge loss given in Equation 9:

$$\min (1/2) \|w\|^2 + C \sum_i \xi_i, \quad \text{s.t. } y_i (w^T x_i + b) \geq 1 - \xi_i \quad (9)$$

Kernel evaluation used the radial basis function defined in Equation 10:

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2), \\ \gamma = 1 / (n_{\text{features}} \times \text{Var}(X)) \quad (10)$$

Multilayer Perceptron (MLP) comprised two fully connected hidden layers (100 and 50 neurons) with ReLU activation, as defined in Equation 11:

$$a_j^l = \max(0, \sum_i w_{ij}^l a_i^{l-1} + b_i^l) \quad (11)$$

Training used the Adam optimiser ($\alpha = 0.001$, $\text{max_iter} = 500$) with a softmax output layer producing class probability estimates.

3.4 Validation strategy

Model generalisation performance was assessed using 10-times Repeated Stratified 5-Fold Cross-Validation (10×5 RSKF) applied

exclusively to the training partition. The complete cross-validation procedure is specified in Algorithm 3. This procedure generates 50 independent accuracy estimates per classifier, enabling reliable estimation of both mean generalisation performance and its variance, as given in Equation 12:

$$CV = (1/kr) \sum_i s_i \pm \sqrt{\left(1/(kr-1)\right) \sum_i (s_i - CV)^2} \quad (12)$$

where $k = 5$ is the number of folds, $r = 10$ is the number of repetitions, and s_i denotes the performance score on fold i . Stratification within each fold preserves class distributions; repeated evaluation with different random seeds produces more stable uncertainty estimates than single-run cross-validation.

The choice of 10 repetitions combined with 5 folds—yielding 50 accuracy estimates per classifier—is motivated by two complementary considerations. A single 5-fold evaluation produces estimates whose variance is strongly influenced by the particular random partition selected: a lucky or unlucky fold split can artificially inflate or deflate the apparent performance of a given classifier. Repeating the process with 10 different random seeds effectively averages over this partition-specific variance, producing a standard deviation estimate that reflects the true variability of out-of-sample performance rather than the noise of a single partitioning decision. Five folds were chosen over ten because each fold allocates 80% of the training data for fitting and 20% for validation; this allocation is particularly important for high-variance ensemble models like Random Forest, whose performance improves substantially with training set size. The 50-score matrix generated by this procedure also constitutes the input to the Friedman statistical test, where a larger number of observations per classifier increases the statistical power of the non-parametric test to detect

ALGORITHM 3 10×5 repeated stratified K-fold cross validation.

```

Input: Training set  $X_{\text{train}}, Y_{\text{train}}$ ; classifier set  $C = \{\text{LR, RF, GB, SVM-RBF, MLP}\}$ ;  $k = 5$  folds;
 $R = 10$  repetitions

Output: Score matrix  $S$  ( $50 \times |C|$ ); mean CV accuracy and std per classifier

1 Initialise score matrix  $S = \text{zeros}(50, |C|)$ 
2 for  $r = 1$  to  $R = 10$  do
2.1 Set random seed = base_seed +  $r$  (distinct partition each repetition)
2.2 Partition  $X_{\text{train}}$  into  $k=5$  stratified folds  $F_1, \dots, F_5$ 
2.3 for fold  $f = 1$  to  $k = 5$  do
     $X_{\text{val}} = F_f$ ;  $X_{\text{fit}} = X_{\text{train}} \setminus F_f$ 
    for each classifier  $C_j$  do
        Train  $C_j$  on  $X_{\text{fit}}$ 
         $\hat{Y}_{\text{pred}} = C_j.\text{predict}(X_{\text{val}})$ 
         $s = \text{accuracy}(\hat{Y}_{\text{pred}}, Y_{\text{val}})$ 
     $S[(r-1) \cdot k + f, j] \leftarrow s$ 
3 for each classifier  $C_j$  do
     $f$   $CV_{\text{mean}}[j] = \frac{1}{50} \cdot \sum S[:, j]$ 
     $f$   $CV_{\text{std}}[j] = \sqrt{\frac{1}{49} \cdot \sum (S[:, j] - CV_{\text{mean}}[j])^2}$ 
4 return  $S, CV_{\text{mean}}, CV_{\text{std}}$ 

```

genuine performance differences that a smaller score matrix would miss.

3.5 Evaluation metrics

A multi-metric evaluation strategy was employed across eight metrics. The weighted F1-score accounts for class frequency when aggregating class-specific F1 scores, as defined in Equation 13:

$$F_1^{\text{Weighted}} = \sum^k (|C^k|/n) \cdot (2 \cdot \text{Precision}^k \cdot \text{Recall}^k) / (\text{Precision}^k + \text{Recall}^k) \quad (13)$$

The Matthews Correlation Coefficient (MCC) measures the correlation between observed and predicted classifications across all classes, as expressed in Equation 14:

$$\text{MCC} = (\text{TP} \cdot \text{TN} - \text{FP} \cdot \text{FN}) / \sqrt{((\text{TP} + \text{FP})(\text{TP} + \text{FN}))((\text{TN} + \text{FP})(\text{TN} + \text{FN}))} \quad (14)$$

AUC-ROC was computed under the one-vs-rest decomposition with weighted averaging. Together, the eight metrics—accuracy, precision, recall, F1-score, AUC-ROC, MCC, Cohen's Kappa, and log loss—provide a comprehensive characterisation of classifier performance under multiclass conditions.

3.6 Ablation study design

To quantify the contribution of each preprocessing component, a structured ablation study was conducted on Dataset 1 using Random Forest as the reference classifier. Four experimental configurations were evaluated:

Configuration 1 (Full pipeline): SMOTE + z-score normalisation + 10×5 RSKF cross-validation.

Configuration 2 (Without SMOTE): Training on the imbalanced partition without synthetic oversampling, with 10×5 RSKF applied to the original training distribution.

Configuration 3 (Without normalisation): SMOTE applied, but z-score standardisation omitted. For Random Forest applied to the binary features of Dataset 1, this configuration is scale-invariant by construction, providing an explicit confirmation that the Gini criterion is unaffected by monotone feature transformations.

Configuration 4 (Single 5-fold CV): Full pipeline applied, but cross-validation reduced to a single run of 5-fold without repetition, enabling direct comparison of variance estimates under repeated versus single-run protocols.

3.7 Statistical significance testing

To determine whether the observed performance differences among the five classifiers are statistically significant or attributable to sampling variability, a non-parametric Friedman test was applied to the 50 cross-validation accuracy scores per model. The Friedman statistic is computed as shown in Equation 15:

$$\chi^2_F = [12n / (k(k+1))] \sum_j (R_j - (k+1)/2)^2 \quad (15)$$

where n is the number of cross-validation folds, k is the number of classifiers, and R_j is the mean rank of classifier j across all folds. Rejection of the null hypothesis of identical classifier distributions ($\alpha = 0.05$) triggers post-hoc Nemenyi pairwise analysis, which determines the critical difference (CD) at a given significance level, as defined in Equation 16:

$$\text{CD} = q\alpha \sqrt{[k(k+1)/6n]} \quad (16)$$

where $q\alpha$ is the critical value from the Studentised range distribution. Two classifiers are declared to differ significantly when their mean rank difference exceeds CD.

The Friedman test was selected over parametric alternatives—most notably a repeated-measures ANOVA—for two reasons rooted in the specific properties of cross-validation accuracy scores. First, accuracy scores are bounded in the interval $[0, 1]$ and, particularly when classifiers perform well, tend to exhibit skewed distributions that violate the normality assumption required for ANOVA to be valid. Second, the Friedman test operates on rank-transformed scores rather than raw values, making it robust to outliers that can arise when a specific fold partition is unusually favourable or unfavourable for a given classifier. The rank-based approach is also conceptually consistent with the comparative nature of the question: we seek to determine which classifier performs best across folds, not whether mean accuracy differences exceed a specific threshold in absolute units. The Nemenyi post-hoc test was selected for pairwise comparisons because it controls the family-wise error rate across all pairwise comparisons simultaneously—across ten possible pairs from five classifiers—making it appropriate when all pairwise contrasts are of equal interest, as is the case here.

3.8 Explainability via TreeSHAP

Shapley Additive Explanations (SHAP) were applied to the best-performing classifier using the TreeSHAP algorithm, which computes exact Shapley values for tree-based models in polynomial time (Liu and Yu, 2026). The complete feature-attribution procedure is specified in Algorithm 4. The Shapley value ϕ_i for feature i quantifies its marginal contribution to a specific prediction, averaged over all possible feature orderings, as given in Equation 17:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F|-|S|-1)!/|F|!}{[f(S \cup \{i\}) - f(S)]} \quad (17)$$

where F is the full feature set, S is a subset of features excluding i , and $f(S)$ is the model prediction using only features in S . SHAP values were computed separately for each target class, with global feature importance derived by averaging absolute SHAP values across all observations and classes.

TreeSHAP was selected as the explainability method over several alternatives—including LIME, gradient-based saliency maps, and permutation importance—for reasons that are both theoretical and practical. From a theoretical standpoint, SHAP values satisfy three axiomatic properties that alternative methods do not simultaneously guarantee: efficiency (the sum of SHAP values for all features equals the difference between the model prediction and the global expected prediction),

ALGORITHM 4 Treeshap feature attribution analysis.

```

Input:  Fitted Random Forest  $M$  ; test set  $X_{\text{test}}$  ( $n=220$ ,  $p=20$  features) ; classes  $Y = \{\text{Low, Medium, High}\}$ 

Output: SHAP tensor  $\Phi$  ( $n \times p \times |Y|$ ) ; global feature importance ranking ; per-class beeswarm plots

1 Initialise  $\phi = \text{zeros}(n=220, p=20, |Y|=3)$ 
2 for each test observation  $x_i$ ,  $i = 1 \dots 220$  do
2.1 for each class  $c \in \{\text{Low, Medium, High}\}$  do
    for each tree  $t \in \text{forest } M$  (100 trees) do
        Traverse tree  $t$  with  $x_i$  ; collect decision path
        Compute per-node  $\phi$  contributions via TreeSHAP algorithm
     $\phi[:, c] \leftarrow \sum_{t=1}^{100} \text{contribution}_t(x_i, c)$ 
2.2  $\phi[i, :, c] \leftarrow \phi[:, c]$ 
-----
3 for each feature  $j = 1$  to  $p = 20$  do
    Importance[j] =  $1/n \cdot \sum_i \sum_c |\phi[i, j, c]|$ 
4 Sort by Importance[j] descending ; return top-10 ranking
-----
5 for each class  $c \in \{\text{Low, Medium, High}\}$  do
    Generate beeswarm plot of  $\phi[:, :, c]$  :
    x    x-axis: SHAP value (positive  $\Rightarrow$  pushes toward class  $c$ )
    y    y-axis: features sorted by mean  $|\text{SHAP}|$ 
    c    colour: feature value (blue = low – red = high)
-----
6 return  $\Phi$ , global feature ranking, per-class beeswarm plots

```

symmetry (features with identical contributions receive identical SHAP values), and the dummy property (features that contribute nothing to any prediction receive zero SHAP values). These properties make SHAP attributions uniquely consistent and additively decomposable across features and observations. From a practical standpoint, the TreeSHAP algorithm computes exact Shapley values for tree-based models in polynomial time, specifically $O(TLD^2)$ for a forest of T trees with maximum depth D and L leaves, compared to the exponential complexity of naive exact computation. This efficiency makes it feasible to compute instance-level explanations for all 220 test observations without approximation. LIME, by contrast, introduces additional error through its sampling-based local linearisation and produces less stable attributions across repeated calls. Permutation importance captures only global marginal effects and is known to overestimate the importance of correlated features—a relevant concern given the moderate intercorrelations observed among psychological indicators in Dataset 2.

4 Results

This section presents the empirical findings in five stages: exploratory analysis of the feature structure of Dataset 2; comparative evaluation of the five classifiers on the held-out test partition; cross-validation stability assessment; ablation study results; and statistical significance analysis followed by SHAP-based explainability.

4.1 Exploratory data analysis

Prior to model training, the internal structure of Dataset 2 was examined through Spearman rank correlation analysis and inter-group distributional comparisons across the three stress severity classes. The Spearman correlation matrix (Figure 3) reveals that anxiety level,

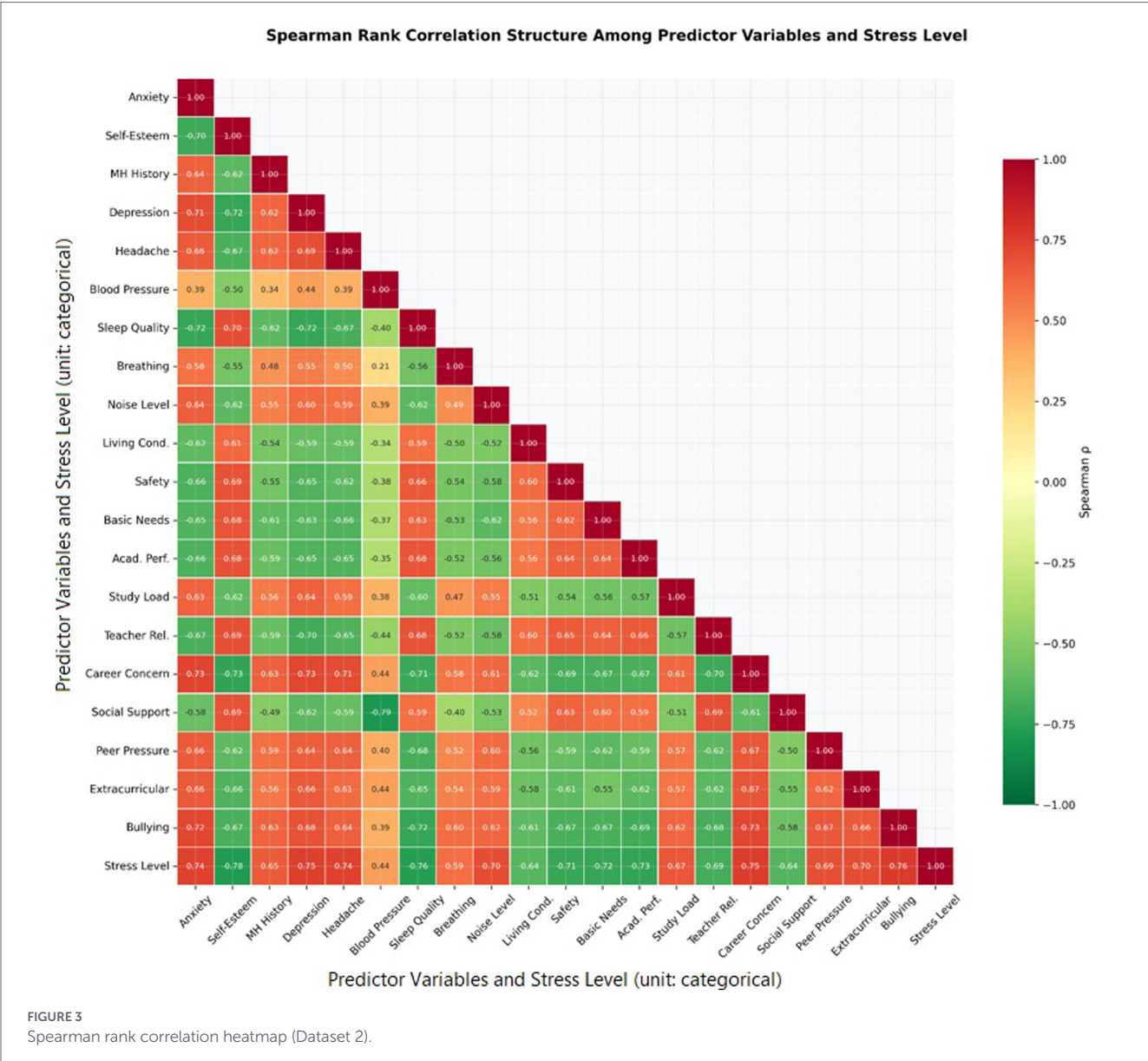
depression, and self-esteem exhibit the strongest correlations with stress level, indicating that psychological indicators are the most linearly associated with stress severity. Self-esteem demonstrates a negative correlation consistent with the established literature linking low self-esteem to heightened vulnerability to academic stress. Sleep quality similarly shows a meaningful negative association, corroborating neurobiological evidence that sleep disruption both results from and exacerbates stress responses. Environmental variables display weaker but directionally consistent associations, suggesting that contextual factors contribute in a less deterministic manner.

The inter-group distributional analysis (Figure 4) confirms that anxiety level, self-esteem, depression, and sleep quality exhibit the most pronounced separation between stress classes, with distributions shifting monotonically across the Low–Medium–High gradient. By contrast, extracurricular activities and safety show comparatively overlapping distributions, suggesting limited discriminative power in isolation. These observations motivate the use of multivariate classification models capable of exploiting feature interactions that are not evident in univariate analyses.

These distributional patterns collectively inform the feature selection context for the classification experiments reported in the following subsection.

4.2 Comparative model performance

All five classifiers were evaluated on the held-out test partition of Dataset 2 ($n = 220$) across eight performance metrics. As summarised in Table 4, Random Forest achieved the highest classification accuracy (0.8909), precision (0.8926), F1-score (0.8907), MCC (0.8373), and Cohen's Kappa (0.8364), while also recording the lowest log loss (0.2057), establishing it as the best-performing classifier across the majority of criteria. Despite its comparatively simple linear decision boundary, Logistic Regression demonstrated the highest AUC-ROC



(0.9853), indicating superior discriminative performance in a threshold-independent sense—a result attributable to the substantial linear separability retained within the feature space of Dataset 2.

Gradient Boosting and MLP achieved identical accuracy (0.8727), with GB demonstrating marginally superior F1-score and MCC, and substantially lower log loss (0.3323 vs. 0.5244). The elevated log loss of MLP indicates overconfident incorrect predictions, consistent with the known sensitivity of neural network calibration to dataset size. SVM-RBF occupied an intermediate position, achieving accuracy of 0.8773 and AUC-ROC of 0.9845. The narrow performance spread across all five models—approximately 0.018 in accuracy and 0.003 in AUC-ROC—indicates that the classification signal in Dataset 2 is sufficiently strong to be captured by a range of algorithmic approaches.

As shown in Figure 5, Random Forest leads on four of the five displayed metrics, while Logistic Regression achieves the highest AUC-ROC, consistent with the values reported in Table 4. The discriminative capacity of each classifier across all three stress severity classes is further examined in Figure 6, with AUC values ranging from 0.9825 to 0.9853—a narrow spread indicating that all models retain strong rank-ordering

ability despite differences in calibration. The class-level prediction patterns are examined in Figure 7, which confirms that misclassifications are distributed across adjacent severity classes rather than concentrated in any single category, suggesting that prediction errors reflect borderline cases rather than systematic failures.

Taken together, Figures 5–7 confirm that Random Forest provides the most consistent classification behaviour across metrics, classes, and decision thresholds, while all five models demonstrate acceptable performance on the balanced Dataset 2.

4.3 Cross-validation stability and learning behaviour

To characterise the stability and generalisation reliability of each classifier, 10×5 RSKF was applied to the training partition, yielding 50 independent accuracy estimates per model. Table 5 reports the resulting means and standard deviations, while Figures 8, 9 illustrate the distributional spread of cross-validation scores and the bias–variance behaviour of each classifier across increasing training set sizes.



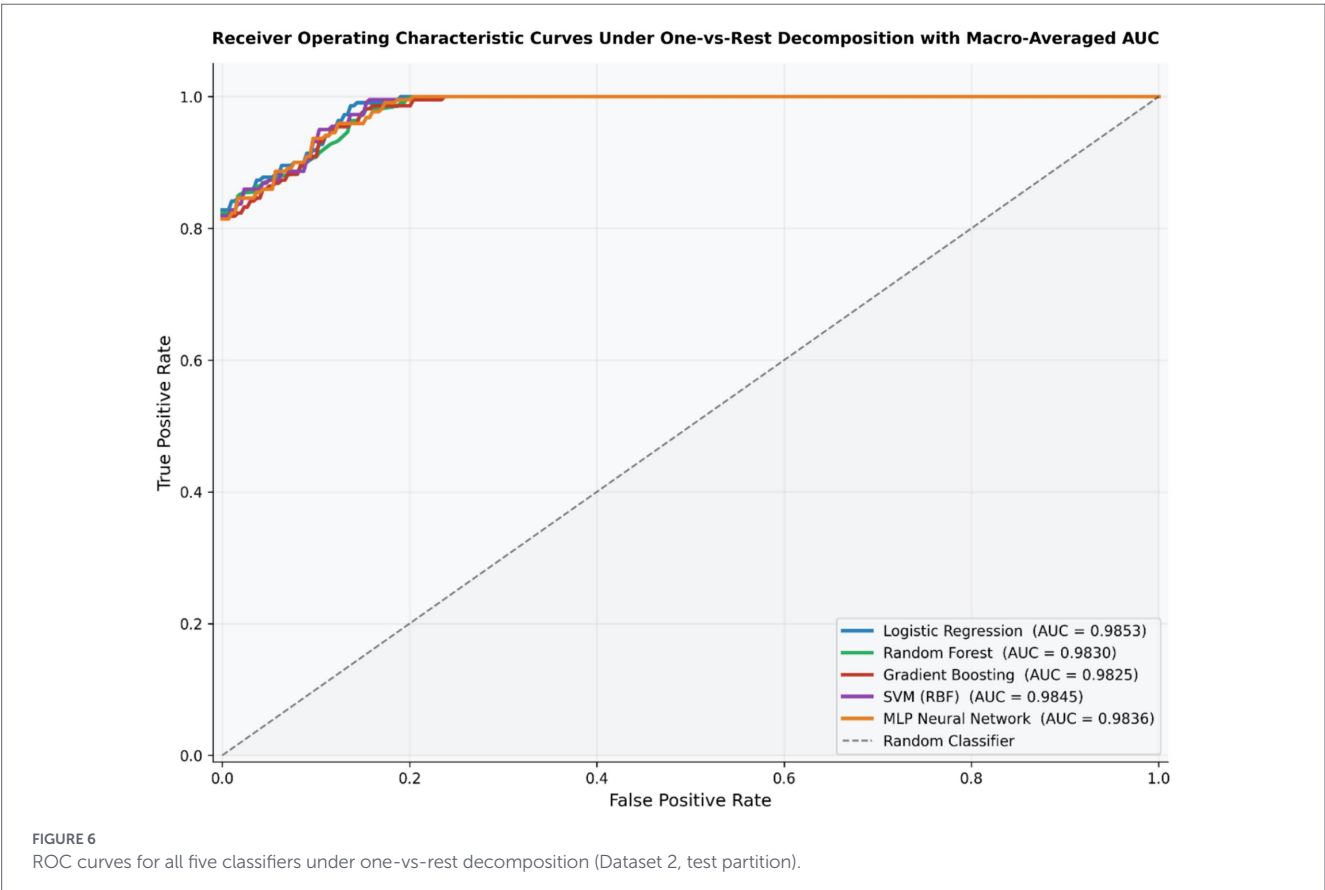
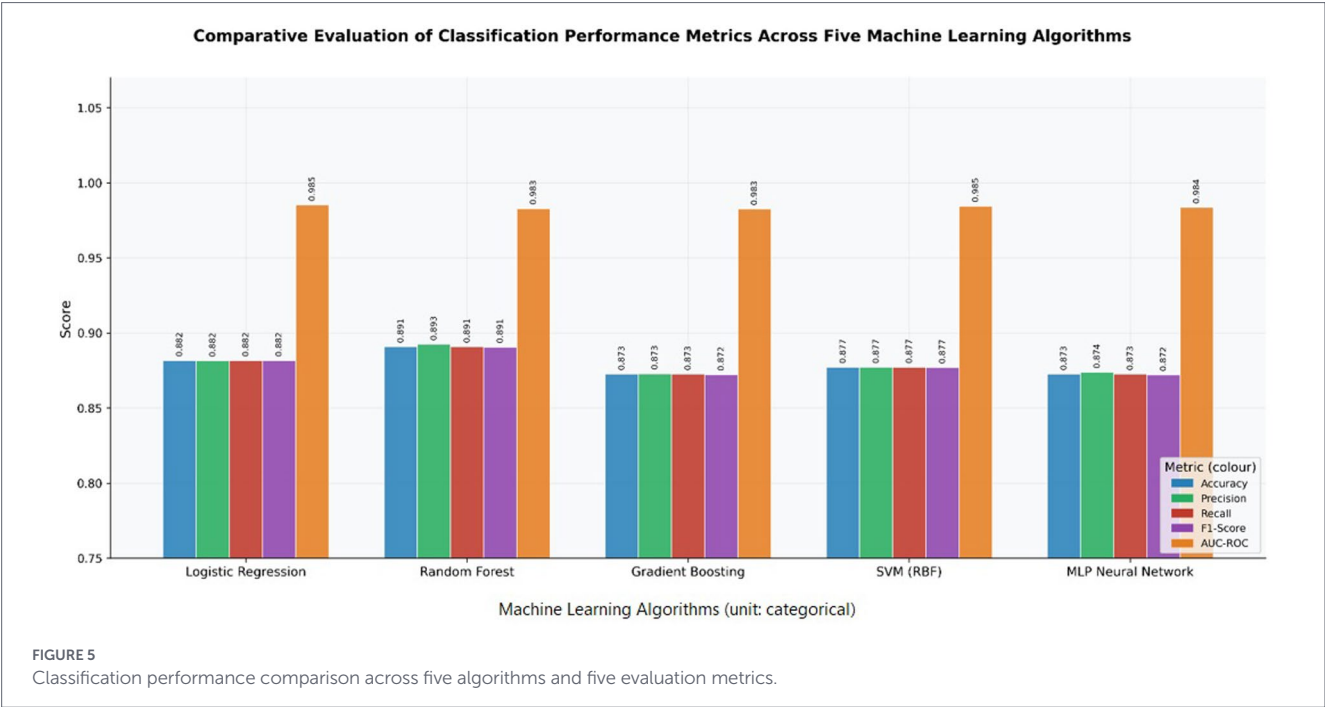
TABLE 4 Comparative performance of the five classification algorithms on the held-out test partition of Dataset 2 ($n = 220$).

Model	Acc	Prec	Rec	F1	AUC-ROC	MCC	Kappa	Log loss
Logistic Regression	0.8818	0.8816	0.8818	0.8817	0.9853	0.8227	0.8227	0.2930
Random Forest	0.8909	0.8926	0.8909	0.8907	0.9830	0.8373	0.8364	0.2057
Gradient Boosting	0.8727	0.8730	0.8727	0.8722	0.9825	0.8098	0.8091	0.3323
SVM (RBF)	0.8773	0.8773	0.8773	0.8771	0.9845	0.8161	0.8159	0.2752
MLP Neural Network	0.8727	0.8739	0.8727	0.8721	0.9836	0.8102	0.8092	0.5244

Acc, Accuracy; Prec, Precision; Rec, Recall; AUC = AUC-ROC. All models trained with `random_state = 42`. Test set results are single held-out partition estimates ($n = 220$); confidence intervals are reported in Table 5 via 10×5 RSKE. Bold values indicate the best result per metric.

All models exhibit standard deviation values in the range 0.0222–0.0235, confirming high stability across the 50 cross-validation folds. Random Forest exhibits the largest positive divergence (+0.0186), indicating that it benefits most substantially from the full training partition—a characteristic of high-variance ensemble methods that improve markedly as training sample size increases. Logistic Regression shows near-zero divergence (+0.0012), consistent with the stable, low-variance

behaviour of a regularised linear model. Gradient Boosting is the only model where test accuracy falls marginally below CV mean (−0.0056), suggesting slight sensitivity to the specific test partition composition. As shown in Figures 8, 9, all five classifiers generalise consistently, with Random Forest showing the greatest capacity to exploit larger training partitions and Logistic Regression demonstrating the most stable bias–variance profile.



4.4 Ablation study results

Table 6, Figure 10 report the outcome of four experimental configurations evaluated on Dataset 1 using Random Forest as the reference classifier. The most consequential finding concerns the divergence between aggregate accuracy and per-class recall: all four configurations achieve identical test accuracy (0.9359), yet this figure

conceals a dramatic collapse in minority-class performance when SMOTE is removed.

The ablation results reveal an important distinction between aggregate accuracy and per-class recall. All four configurations achieve identical test accuracy (0.9359), reflecting the dominance of the Eustress majority class in the test partition. However, the per-class recall analysis—presented in Figure 10B—demonstrates

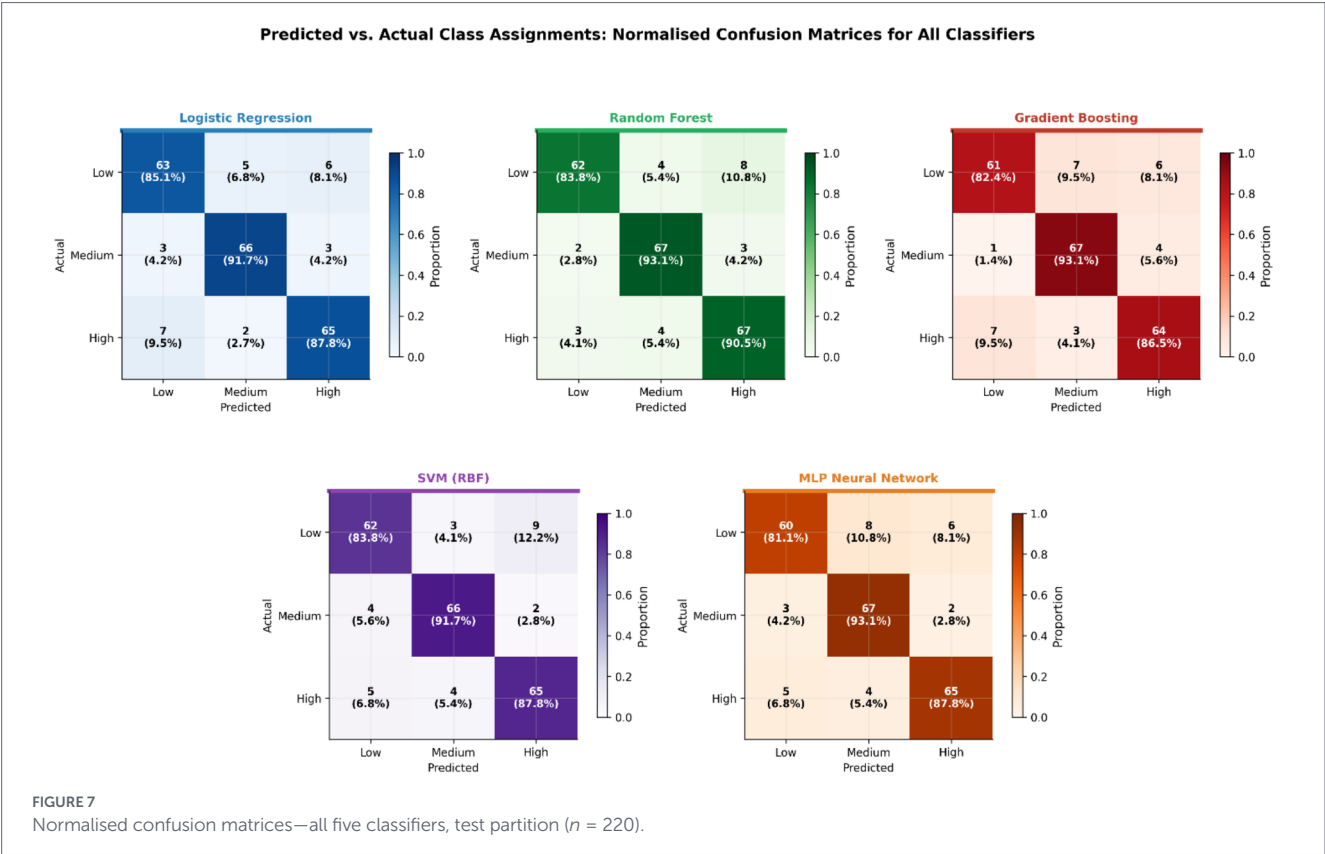


TABLE 5 Cross-validation results (10 × 5 RSKF) for all five classifiers on the training partition of Dataset 2.

Model	CV Acc Mean	CV Acc Std	CV Acc Best	Test Acc	Δ (Test – CV Mean)
Logistic Regression	0.8806	0.0224	0.9227	0.8818	+0.0012
Random Forest	0.8723	0.0224	0.9545	0.8909	+0.0186
Gradient Boosting	0.8783	0.0232	0.9318	0.8727	−0.0056
SVM (RBF)	0.8636	0.0235	0.9227	0.8773	+0.0137
MLP Neural Network	0.8690	0.0222	0.9318	0.8727	+0.0037

The Δ column denotes the difference between test set accuracy and mean CV accuracy.

a stark divergence. Without SMOTE (Configuration 2), recall for Distress collapses to 0.6667 and recall for No Stress falls to 0.1250, despite the unchanged aggregate accuracy. This confirms that aggregate accuracy is a misleading performance measure under class imbalance: the full pipeline with SMOTE is the only configuration that achieves adequate recall across all three stress type categories.

The comparison of cross-validation variance between Configurations 1 and 4 reveals that without SMOTE, variance increases from 0.0041 to 0.0122—a 198% increase. This is attributable to the sensitivity of minority-class classification to the specific composition of each training fold under the original imbalanced distribution. Notably, single 5-fold CV yields a lower standard deviation (0.0023) than 10 × 5 RSKF (0.0041); this counterintuitive finding reflects the reduced variability across a smaller number of fold evaluations, which underestimates the true variance in generalisation performance. The 10 × 5 RSKF protocol, by sampling 50 independent fold partitions, produces a more reliable and conservative estimate of out-of-sample performance variability.

As shown in Figure 10, SMOTE is an essential component of the pipeline for any application where minority-class recall carries clinical or institutional significance, and 10 × 5 RSKF provides a more conservative and reliable variance estimate than single-run cross-validation.

4.5 Statistical significance testing

The Friedman test was applied to the 50 cross-validation accuracy scores per classifier, yielding a test statistic of $\chi^2 = 68.97$ ($p < 0.001$). This result rejects the null hypothesis of equivalent classifier performance distributions, confirming that the observed performance differences are statistically significant and not attributable to sampling variability. *Post-hoc* Nemenyi analysis was conducted to identify which specific pairs of classifiers differed significantly. The resulting pairwise p -values, detailed in Table 7 and illustrated as a heatmap in Figure 11, reveal a structured two-cluster pattern: Random Forest and Logistic Regression form a statistically superior group, while Gradient Boosting, SVM-RBF, and MLP constitute a lower-performing cluster, with the boundary between clusters confirmed at $\alpha = 0.05$.

Stability of Generalisation Performance Under 10-Times Repeated Stratified 5-Fold Cross-Validation

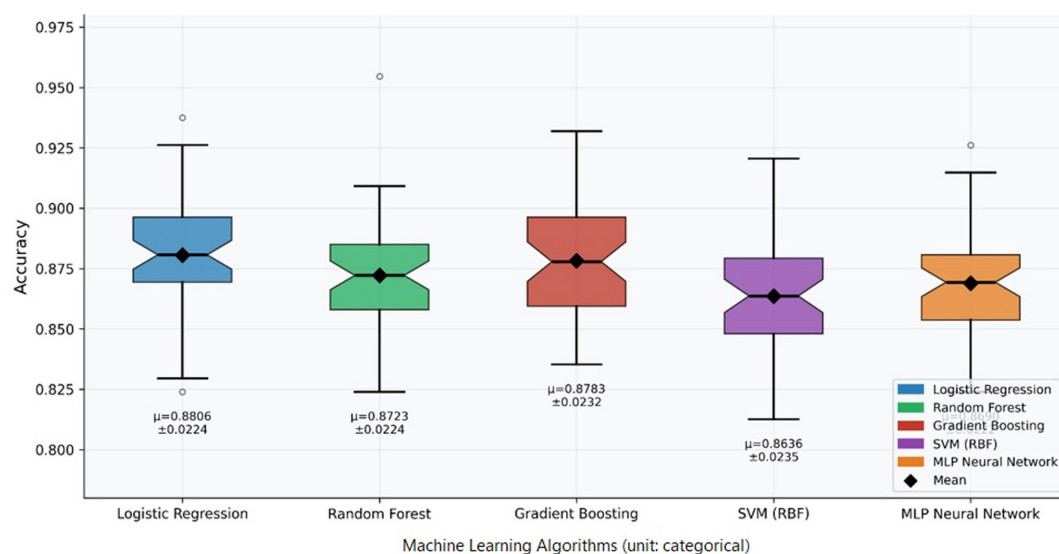


FIGURE 8

Cross-validation accuracy distributions across 50 folds for all five classifiers.

Bias-Variance Characterisation Through Training and Validation Learning Trajectories

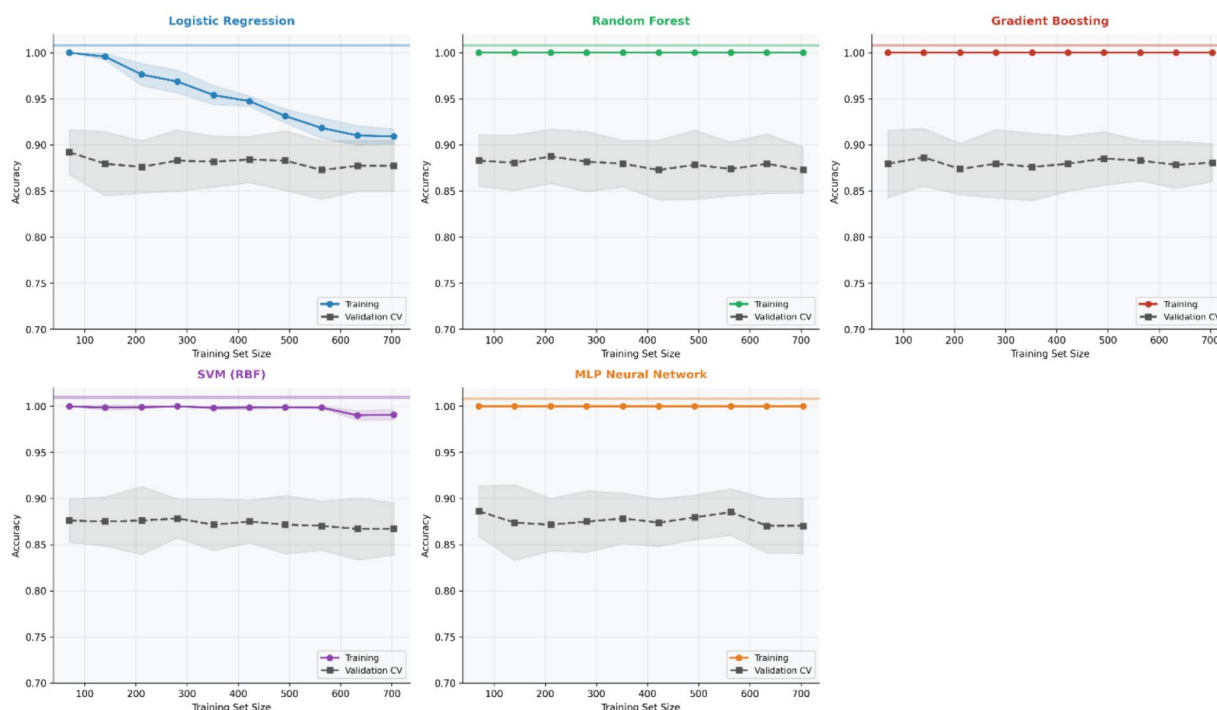


FIGURE 9

Learning curves—bias—variance characterisation, all five classifiers.

The Nemenyi analysis reveals a structured pattern of pairwise significance across all ten possible comparisons, visualised in Figure 11. Random Forest significantly outperforms both Logistic Regression ($p = 0.0051$) and SVM-RBF ($p = 0.0020$), but does not significantly differ from Gradient Boosting ($p = 0.1989$) or MLP ($p = 0.6127$). Logistic Regression significantly outperforms SVM-RBF ($p < 0.0001$)—the largest pairwise difference observed—and also significantly outperforms MLP ($p < 0.0001$), while not differing significantly from Gradient

Boosting ($p = 0.6934$). Within the lower-performing group, Gradient Boosting significantly outperforms both MLP ($p = 0.0032$) and SVM-RBF ($p < 0.0001$), whereas SVM-RBF and MLP do not differ significantly from each other ($p = 0.1524$). These results define two performance clusters: a higher-performing group comprising Random Forest and Logistic Regression, and a lower-performing group comprising Gradient Boosting, SVM-RBF, and MLP. Within the lower cluster, Gradient Boosting occupies a statistically distinct position above

TABLE 6 Ablation study results—Dataset 1, Random Forest classifier.

Configuration	Test accuracy	Weighted F1	Δ F1 vs. Full	CV Std
Full pipeline (proposed)	0.9359	0.9227	–	0.0041
Without SMOTE	0.9359	0.9185	–0.0042	0.0122
Without normalisation	0.9359	0.9227	0.0000	0.0041
Single CV (no repetition)	0.9359	0.9227	0.0000	0.0023

Identical test accuracy across all configurations is attributable to the 91.4% Eustress majority class in the test partition; per-class recall differences are reported in Figure 10B. Test accuracy is identical across all configurations (0.9359), reflecting majority-class dominance; weighted F1 and CV Std reveal meaningful differences.

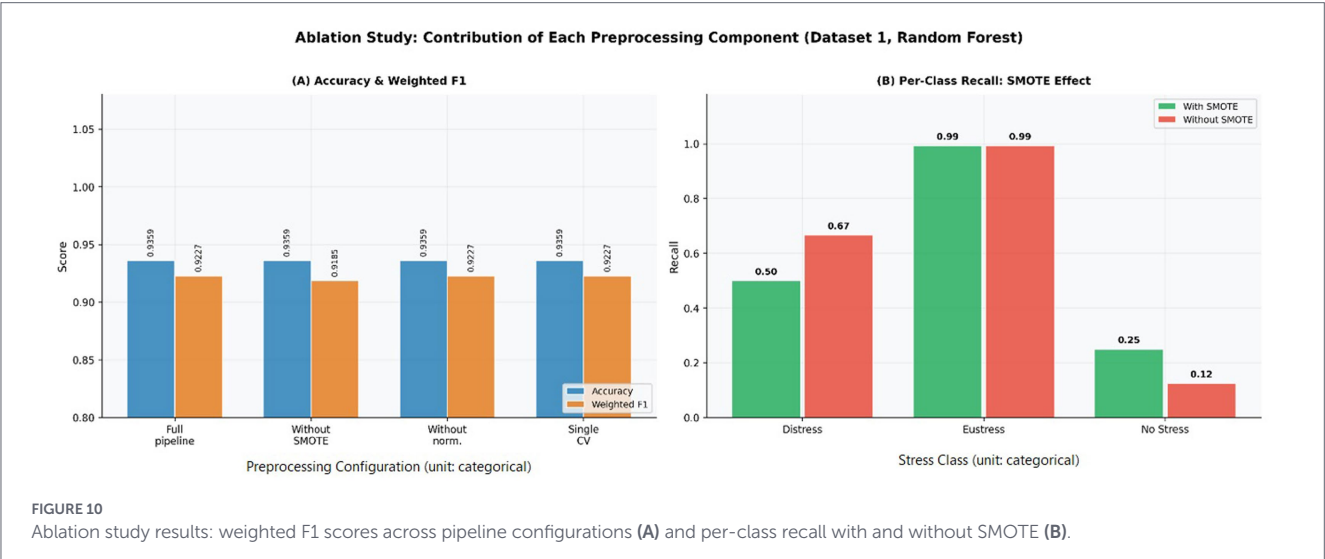


TABLE 7 Post-hoc Nemenyi pairwise significance results following Friedman test ($\chi^2 = 68.97, p < 0.001$).

Comparison	p-value	Significant ($\alpha = 0.05$)?
Random Forest vs. Logistic Regression	0.0051	Yes
Random Forest vs. Gradient Boosting	0.1989	No
Random Forest vs. SVM (RBF)	0.0020	Yes
Random Forest vs. MLP Neural Network	0.6127	No
Logistic Regression vs. Gradient Boosting	0.6934	No
Logistic Regression vs. SVM (RBF)	<0.0001	Yes
Logistic Regression vs. MLP Neural Network	<0.0001	Yes
Gradient Boosting vs. SVM (RBF)	<0.0001	Yes
Gradient Boosting vs. MLP Neural Network	0.0032	Yes
SVM (RBF) vs. MLP Neural Network	0.1524	No

Friedman test applied to 50 cross-validation accuracy scores per classifier.

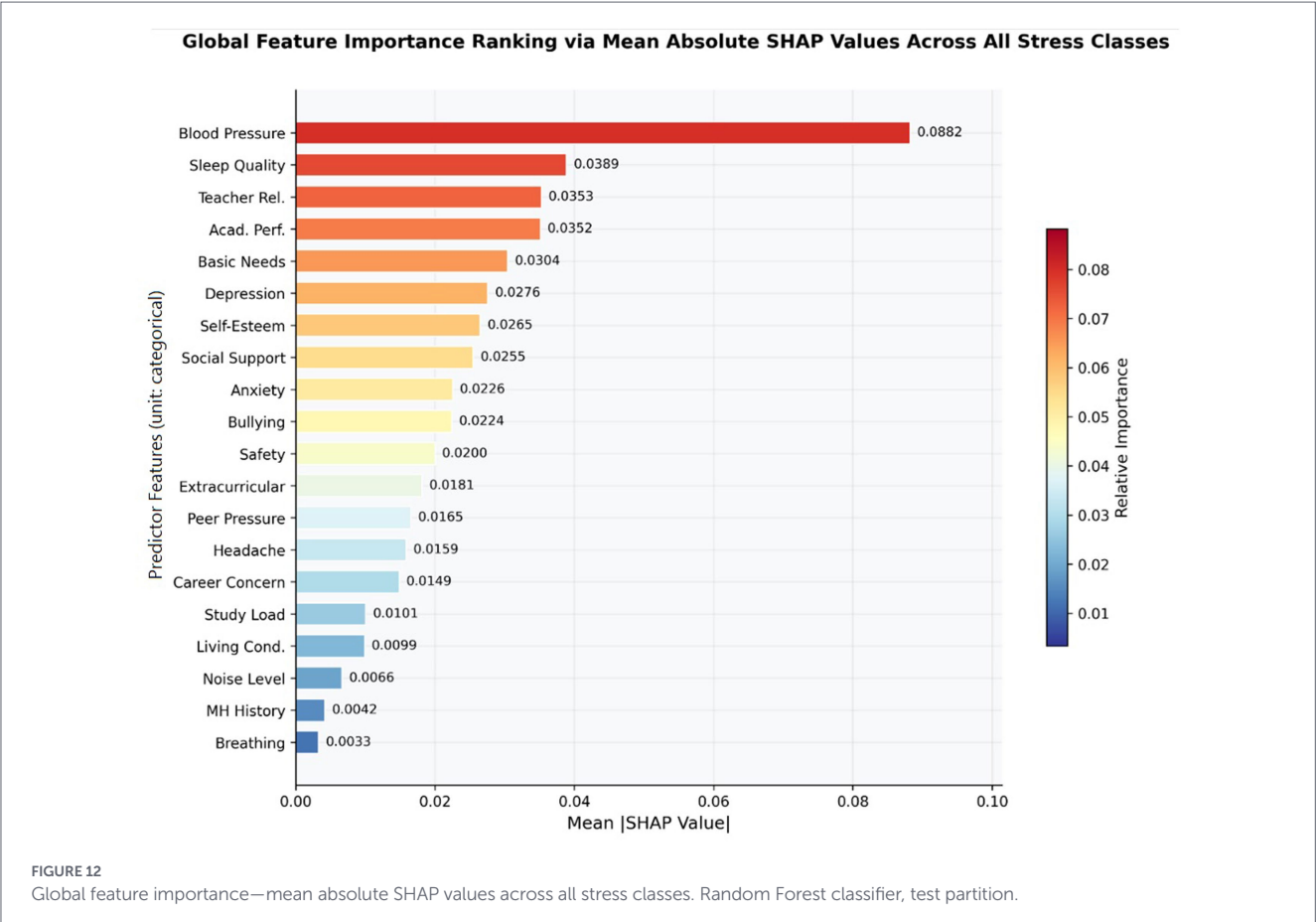
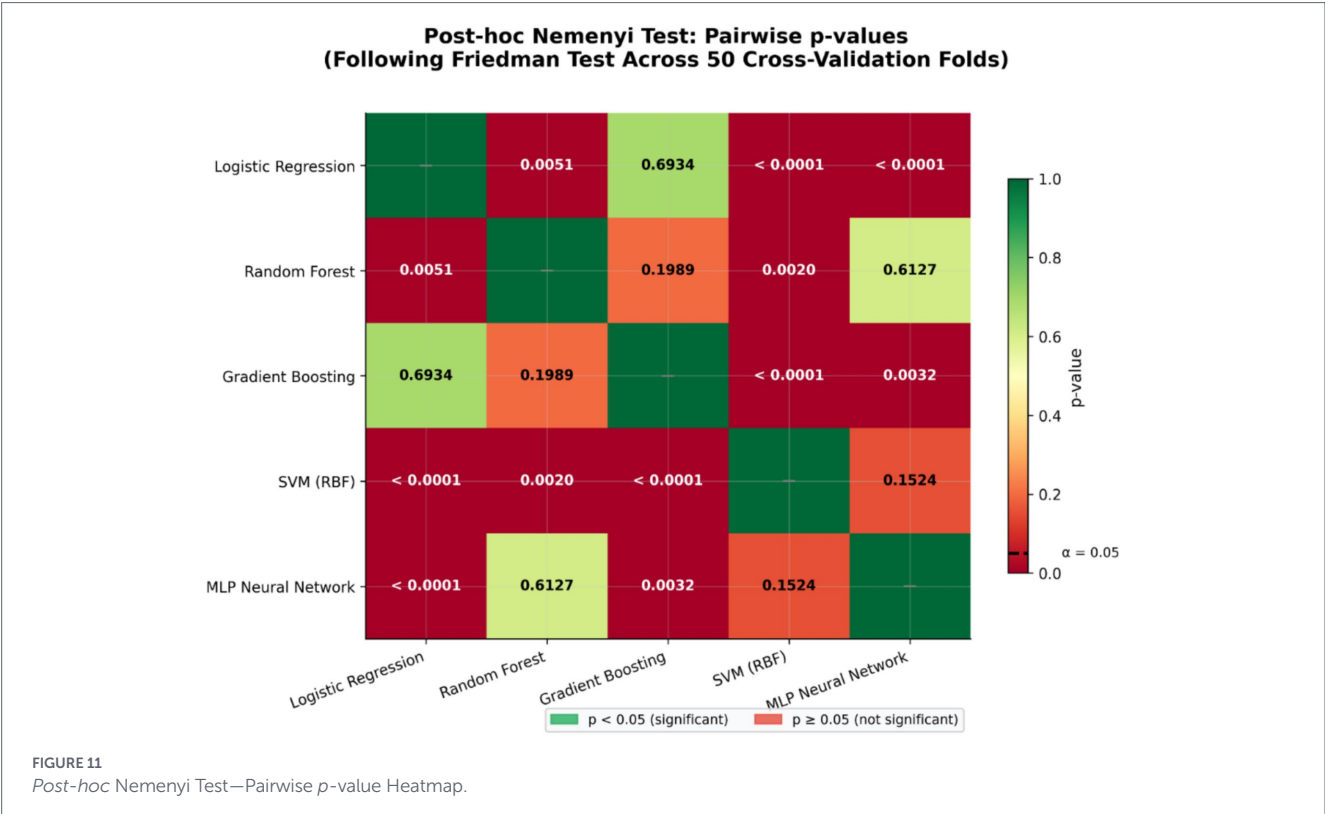
SVM-RBF and MLP, introducing a secondary stratification that the aggregate cluster description does not fully capture.

It should be noted that the classification of classifiers into two performance clusters is based on the pattern of statistically significant pairwise differences identified by the Nemenyi *post-hoc* test, rather than on an arbitrary threshold applied to raw accuracy values. Specifically, both Random Forest and Logistic Regression significantly outperform SVM-RBF and MLP ($p < 0.05$), while not significantly differing from each other ($p = 0.0051$), justifying their grouping as a statistically coherent superior cluster. Within the lower-performing group, Gradient Boosting occupies a statistically distinct position above both SVM-RBF and MLP, representing a secondary stratification within the cluster.

As shown in Figure 11, the two-cluster structure is visually apparent, confirming that the performance advantage of Random Forest and Logistic Regression over the remaining classifiers is a statistically robust finding across 50 independent fold evaluations.

4.6 SHAP explainability analysis

SHAP-based explainability analysis was applied to Random Forest. TreeSHAP was used to compute exact Shapley values for all 220 test observations across all three stress classes. The global feature importance ranking presented in Figure 12 identifies blood pressure as the most influential predictor of stress severity across all three stress classes (mean $|\text{SHAP}| = 0.0882$), followed by sleep quality (0.0389),



teacher–student relationship (0.0353), and academic performance (0.0352). Table 8 reports the global feature importance ranking averaged across all three stress classes. At the level of the High Stress class specifically, self-esteem ranks first (mean |SHAP| = 0.0583), followed by sleep quality, bullying, and anxiety level—a distribution that differs from the global ranking due to the concentrated predictive weight of self-esteem within the High Stress class.

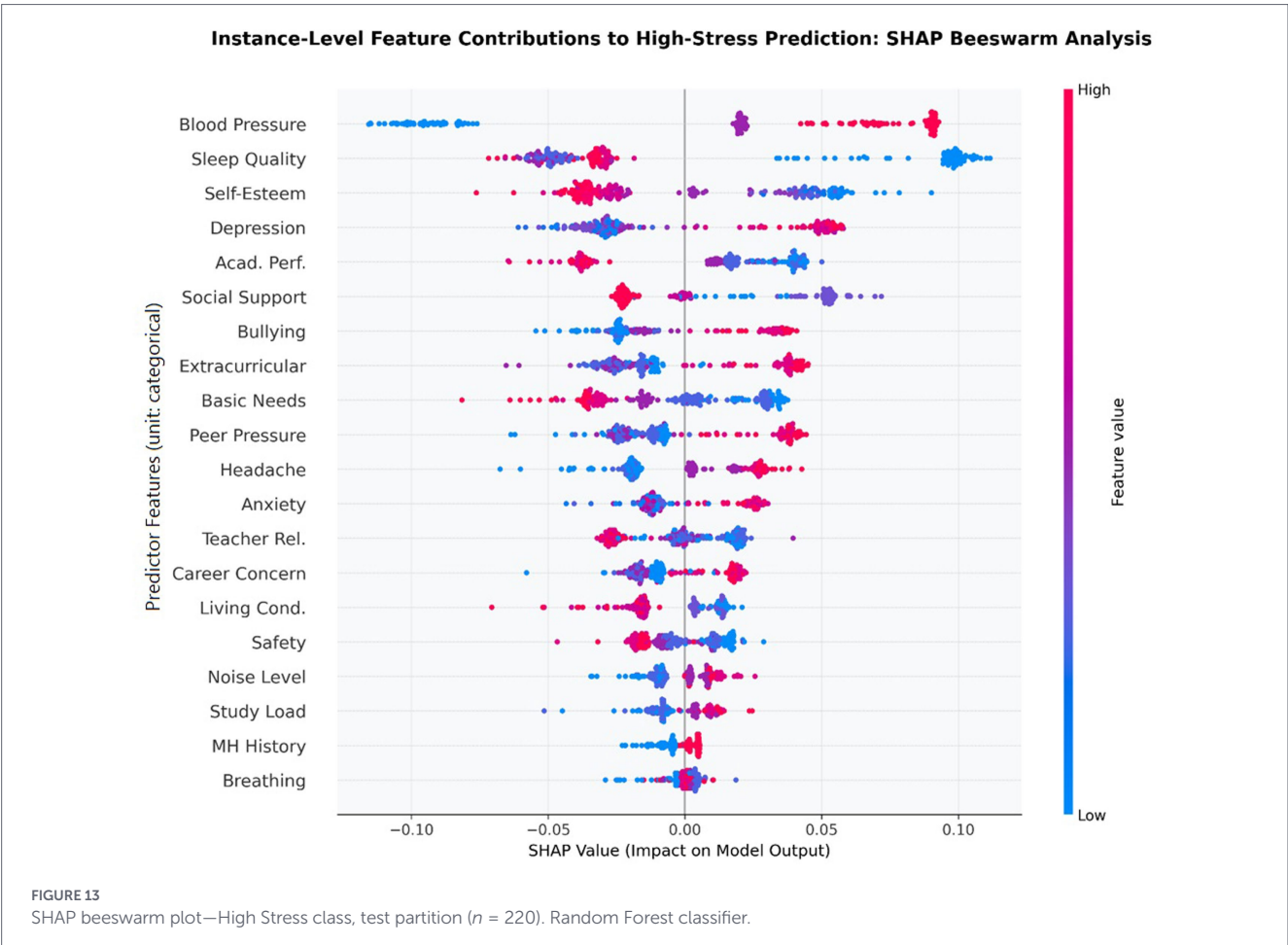
The beeswarm plot for the High Stress class (Figure 13) provides instance-level granularity. High SHAP values for self-esteem are

associated with low feature values—students with low self-esteem receive substantially higher predicted probabilities for the High Stress class. The same directional pattern is observed for sleep quality: poor sleep consistently pushes predictions toward the High Stress category. Bullying exhibits a positive directional effect—higher reported bullying scores are associated with increased High Stress predictions—confirming that peer victimisation is a direct driver of severe stress classification.

As shown in Figures 12, 13, the convergence of global and class-specific SHAP rankings around psychological and physiological

TABLE 8 Top-10 predictor features ranked by mean absolute SHAP value—global ranking across all stress classes (Random Forest, test partition, $n = 220$).

Rank	Feature	Domain	Mean SHAP	Direction (high stress)
1	Blood pressure	Physiological	0.0882	Positive ↑
2	Sleep quality	Physiological	0.0389	Negative ↓
3	Teacher relationship	Environmental	0.0353	Negative ↓
4	Academic performance	Academic	0.0352	Negative ↓
5	Basic needs	Environmental	0.0304	Negative ↓
6	Depression	Psychological	0.0276	Positive ↑
7	Self-esteem	Psychological	0.0265	Negative ↓
8	Social support	Social	0.0255	Negative ↓
9	Anxiety	Psychological	0.0226	Positive ↑
10	Bullying	Social	0.0224	Positive ↑



predictors reinforces the multidimensional character of academic stress and provides actionable targets for institutional intervention programmes.

5 Discussion

The comparative evaluation of five classification algorithms yielded consistently strong performance, with accuracy values ranging from 0.8727 to 0.8909 and AUC-ROC values from 0.9825 to 0.9853. The superiority of Random Forest across the majority of criteria is consistent with the established literature demonstrating that ensemble tree methods achieve competitive performance on structured tabular educational datasets (Fonda et al., 2024; Ratul et al., 2023; Tariq et al., 2025). The comparatively low log loss of RF (0.2057 vs. 0.5244 for MLP) indicates superior probability calibration—a property of particular importance for institutional deployment, where prediction confidence may influence the urgency of a counselling referral. The finding that Logistic Regression achieved the highest AUC-ROC (0.9853) despite its linear decision boundary suggests that class boundaries in the feature space of Dataset 2 retain substantial linear separability. In contexts where model transparency and regulatory compliance are prioritised, this supports Logistic Regression as a viable alternative to ensemble methods, offering near-equivalent discriminative performance with considerably greater coefficient interpretability.

The ablation study established that SMOTE has no measurable effect on aggregate accuracy but a dramatic effect on per-class recall: without it, recall for No Stress collapses to 0.1250 and Distress recall falls to 0.6667, despite unchanged aggregate accuracy of 0.9359. This confirms that aggregate accuracy is a systematically misleading metric under class imbalance, and minority-class recall is the appropriate criterion for evaluating the clinical utility of a stress classification system. The Friedman test ($\chi^2 = 68.97$, $p < 0.001$) further confirms that performance differences across classifiers are statistically significant rather than attributable to sampling variability. The *post-hoc* Nemenyi analysis defines two distinct performance clusters—RF and LR as the statistically superior group, and GB, SVM-RBF, and MLP as the lower-performing group—with direct practical implications for classifier selection under real deployment constraints. Notably, SVM-RBF achieved respectable test accuracy (0.8773) yet ranked significantly below both leading models in cross-validation, illustrating that single-partition results can overstate generalisation performance in the absence of repeated testing.

Blood pressure emerges as the globally most influential predictor across all three stress severity classes, consistent with the physiological literature linking chronic stress to sustained cardiovascular arousal (Figueroa et al., 2023). At the class-specific level, self-esteem ranks as the primary predictor of High Stress classification—a finding coherent within cognitive appraisal frameworks, where individuals with lower self-esteem evaluate academic demands as threatening rather than challenging (Almarzouki, 2024; Fullerton et al., 2021). Sleep quality ranks second both globally and within the High Stress class, reflecting the well-established bidirectional relationship between sleep disruption and stress activation (Almarzouki, 2024). The emergence of bullying as the third most influential predictor identifies peer victimisation as a major driver of high stress severity independently of academic pressures (Campbell et al., 2022), while the negative directional effects of social support and living conditions

confirm that environmental and social factors contribute independently to severe stress classification. It is important to note that blood pressure in Dataset 2 was measured as a self-reported ordinal rating on a scale of 1–5, reflecting the student's subjective perception of their cardiovascular state rather than a clinical measurement in millimetres of mercury. This distinction is relevant for interpreting its predictive importance: the variable captures the student's awareness of cardiovascular symptoms associated with stress rather than an objective physiological measurement. Furthermore, it must be emphasised that SHAP values quantify the contribution of each feature to the predictions of the trained model and do not constitute evidence of direct causal relationships between predictors and stress outcomes. The high SHAP importance of blood pressure indicates that this self-reported variable is strongly associated with stress severity classification within the model, but causal inference would require controlled experimental or longitudinal study designs that fall outside the scope of the present work.

These methodological observations are further supported by convergent findings in the most recent literature. Tariq et al. (2025) applied explainable AI methods to student stress prediction, corroborating the utility of SHAP-based feature importance analysis for identifying actionable stress predictors. Ahmed et al. (2026) demonstrated that ensemble methods combined with explainability layers achieve strong performance in educational mental health contexts, consistent with the superior performance of Random Forest observed in the present study. The convergence of findings across independently conducted studies using different datasets and methodological configurations strengthens confidence in the robustness of the identified predictors and the validity of the proposed framework.

Addressing the research questions formulated in the introduction: Random Forest provided the most accurate and statistically distinguishable classification of student stress levels, though Logistic Regression is recommended where interpretability is the primary deployment criterion (RQ1). Stress severity is jointly shaped by psychological, physiological, social, and academic factors, with no single domain sufficient for accurate prediction—psychological vulnerability in particular amplifies the effect of environmental and academic stressors (RQ2). Self-esteem, sleep quality, bullying, and anxiety level consistently emerged as the primary drivers of High Stress classification, as identified through SHAP analysis (RQ3). All five models demonstrated high stability across 50 cross-validation folds, and performance differences were confirmed as statistically significant by the Friedman test, with the Nemenyi analysis revealing two distinct performance clusters (RQ4).

The generalisability of the present findings is subject to important contextual boundaries. Both datasets were collected within specific institutional settings, and the predictive relationships identified—particularly the prominence of blood pressure, sleep quality, and self-esteem—may not transfer uniformly across different national educational systems or cultural contexts. External validation on independent datasets from diverse geographical settings is therefore a necessary condition for establishing broader applicability. From a practical implementation perspective, the SHAP-based framework offers a viable foundation for institution-level early warning systems, enabling individual-level risk profiles to be communicated to university counselling services in an interpretable format, supporting targeted intervention rather than population-level screening alone. Responsible deployment would additionally require integration with institutional data governance frameworks, transparent communication to students

regarding data usage, and ongoing monitoring of model fairness across demographic subgroups.

6 Limitations

Several limitations warrant acknowledgement. First, both datasets are based exclusively on self-reported survey responses, which are susceptible to social desirability bias and recall inaccuracies. Second, the geographical and institutional provenance of both datasets is not fully specified, constraining the generalisability of findings to other cultural or educational contexts. Third, all classifiers were evaluated under default hyperparameter configurations. While this ensures a fair baseline comparison unconfounded by optimisation effort, systematic tuning via cross-validated grid or random search may yield further performance improvements, particularly for SVM-RBF and MLP, whose performance is known to be sensitive to kernel bandwidth and learning rate, respectively. Fourth, the cross-sectional design precludes modelling of temporal dynamics in student stress. Fifth, Dataset 1—despite SMOTE augmentation—originates from a severely imbalanced distribution (91.4% Eustress), and the synthetic nature of minority-class samples introduces uncertainty regarding representativeness. Sixth, both datasets are restricted to university students within a narrow age range (18–22 years), which limits the generalisability of the findings to other age groups, educational levels, or non-traditional student populations. The proposed framework operates exclusively on cross-sectional survey data and is therefore not designed for real-time stress monitoring. Deployment in a live institutional setting would require continuous data collection, which introduces additional ethical considerations—including informed consent procedures, data minimisation principles, and compliance with data protection regulations such as GDPR – that fall outside the scope of the current study. Seventh, the absence of external validation on independent datasets from different national, cultural, or institutional contexts represents a meaningful limitation. The predictive relationships identified in the present study—particularly the prominence of blood pressure, sleep quality, and self-esteem as stress predictors—may not generalise uniformly across different educational systems or student populations. Both datasets were collected within a specific geographical and institutional context, and access to independent validated datasets from other national contexts was not available within the scope of the present study. External validation on independent datasets is therefore identified as a necessary and prioritised direction for future research.

7 Conclusion

This study proposed and evaluated a comparative machine learning framework for the classification of academic stress among university students, integrating explainable artificial intelligence, structured ablation analysis, and non-parametric statistical significance testing within a dual-dataset design. Random Forest

achieved the strongest overall classification performance on Dataset 2 (accuracy 0.8909, F1 0.8907, log loss 0.2057), while Logistic Regression demonstrated superior discriminative ability (AUC-ROC 0.9853). The Friedman test ($\chi^2 = 68.97$, $p < 0.001$) confirmed that performance differences are statistically significant, with the Nemenyi post-hoc analysis identifying RF and LR as a statistically superior cluster. The ablation study demonstrated that SMOTE is essential for minority-class recall—its removal reduces No Stress recall to 0.1250 despite unchanged aggregate accuracy—establishing that accuracy alone is an insufficient evaluation criterion under class imbalance. SHAP analysis identified blood pressure, sleep quality, teacher–student relationship, and academic performance as the most globally influential predictors of stress severity, with self-esteem, sleep quality, bullying, and anxiety emerging as the primary drivers of High Stress classification specifically.

These findings carry direct practical implications. Institutional stress monitoring programmes should prioritise self-esteem through counselling initiatives, sleep quality through wellness interventions, and peer victimisation through anti-bullying policies. The SHAP-based framework enables individual-level explanations that can support personalised counselling referrals alongside population-level insights. The statistical validation approach established here—combining Friedman testing with Nemenyi post-hoc analysis—should become standard practice in comparative machine learning studies in educational data mining, replacing informal comparison of point estimates.

Future work should address the limitations of the current study through longitudinal data collection to model temporal dynamics in student stress, incorporation of objective physiological measures to complement self-report data, external validation on independent datasets from different national and cultural contexts, and the development of adaptive real-time monitoring systems capable of providing early warnings before stress escalates to clinically significant levels.

Data availability statement

Dataset 1 (Stress_Dataset.csv) is publicly available at: https://www.kaggle.com/datasets/mdsultanulislamovi/student-stress-monitoring-datasets?select=Stress_Dataset.csv. Dataset 2 (StressLevelDataset.csv) is publicly available at: <https://www.kaggle.com/datasets/samyakb/student-stress-factors>. The complete source code is available at: https://github.com/albina0dali/academic_stress. Both datasets are used exclusively for research purposes in accordance with the respective platform terms of use.

Ethics statement

Ethical approval was not required for the study involving humans in accordance with the local legislation and institutional requirements. Written informed consent to participate in this study was not required from the participants or the participants' legal guardians/next of kin in accordance with the national legislation and the institutional requirements.

Author contributions

AO: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. AN: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. GB: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. ZL: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. AG: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. AA: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that Generative AI was not used in the creation of this manuscript.

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Appendix A

Questionnaire items – dataset 1

All items were rated on a five-point Likert scale ranging from “Not at all” to “Extremely.”

Emotional and Stress Indicators:

1. Have you recently experienced stress in your life?
2. Have you noticed a rapid heartbeat or palpitations?
3. Have you been dealing with anxiety or tension recently?
4. Do you face any sleep problems or difficulties falling asleep?
5. Have you been feeling sadness or low mood?
6. Do you get irritated easily?
7. Do you often feel lonely or isolated?
8. Do you have trouble concentrating on your academic tasks?

Physical and Health Indicators:

9. Have you been getting headaches more often than usual?
10. Have you been experiencing any illness or health issues?
11. Have you gained/lost weight?

Academic and Environmental Stressors:

12. Do you feel overwhelmed with your academic workload?
13. Are you in competition with your peers, and does it affect you?
14. Do you lack confidence in your academic performance?
15. Do you lack confidence in your choice of academic subjects?
16. Academic and extracurricular activities conflicting for you?
17. Do you attend classes regularly?
18. Are you facing any difficulties with your professors or instructors?
19. Is your working environment unpleasant or stressful?
20. Is your hostel or home environment causing you difficulties?

Social and Relationship Factors:

21. Do you find that your relationship often causes you stress?
22. Do you struggle to find time for relaxation and leisure activities?

Target variable: Which type of stress do you primarily experience? (Eustress/Distress/No Stress)